

Restricted Large Bayesian Vector Autoregressions

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Abstract

Without regularization, vector autoregressions (VARs) with many variables suffer from large standard errors and poor out-of-sample forecasting performance. Regularization is typically associated with a bias-variance trade-off, but we show that in finite samples, imposing parameter equality restrictions – akin to clustering parameters along both rows and columns of the coefficient matrix – can reduce bias *and* variance by mitigating second-order bias, even when the restrictions are mildly misspecified. We introduce the Matrix Dirichlet Process (MDP) prior for VARs, which remains agnostic *ex ante* about the appropriate restrictions by jointly sampling the clustering structure and parameter values within a computationally efficient Gibbs sampling framework. Simulations and three applications – forecasting personal income growth across U.S. states, estimating an 18-variable proxy SVAR to identify the effects of monetary policy, and analyzing price spillovers across 24 industries in response to uncertainty shocks – demonstrate that the MDP prior recovers intuitive economic relationships and improves forecasting accuracy.

Keywords: Clustering, large vector autoregression, shrinkage, bias-variance trade-off, Bayesian analysis

JEL classification codes: C11, C22, C53, C55, E37

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1 Introduction

One of the fundamental challenges in modeling economic systems is that as specifications grow richer, capturing interlinkages via systems of equations, the number of parameters in the model increases quadratically with the number of equations. This issue arises in a range of applications, including Vector Autoregressions (VARs), transition probabilities in (hidden) Markov models, network-based models, and when estimating of covariance matrices in multivariate systems. As the number of parameters grows, models suffer from inflated standard errors and poor out-of-sample fit at best, and at worst, they may lose identification entirely as the number of parameters exceeds the number of observations.

To address these challenges, researchers have developed dimension-reduction techniques that impose structure while preserving flexibility. One approach is low-rank factorization, as in Factor-Augmented VARs (FAVARs) (Bernanke, Boivin, and Elias, 2005), which summarize high-dimensional data using a small number of common factors. Alternatively, regularization methods—including LASSO and Ridge regression, Bayesian VARs with Minnesota priors (Litterman, 1979, 1980) or spike-and-slab priors (Giannone, Lenza, and Primiceri, 2021) – address overfitting by shrinking or setting certain coefficients to zero. Economic theory can also motivate dimension-reduction, as in economic networks, imposing factorization or sparsity based on production linkages (Foerster, Sarte, and Watson, 2011; Mlikota, 2023), or spatial econometric models, where entities closer in space are assumed to be more strongly connected.

These methods are often viewed through the lens of a bias-variance trade-off, where reducing variance through additional structure comes at the cost of introducing bias. This paper aims to challenge that common view and suggests that in time series settings, it is possible to think of dimension-reduction methods that simultaneously reduce bias and variance. The key insight is that time series models like VARs often exhibit poor finite-sample properties and suffer from substantial second-order bias, as shown in, among others, Bao and Ullah (2007). We show that imposing parameter equality restrictions – when they are correctly specified – reduces second-order bias, improving inference and estimation uncertainty. Con-

versely, incorrect restrictions introduce misspecification bias. However, as long as the reduction in second-order bias dominates the bias introduced by misspecification, the additional structure from these restrictions can serve as a net improvement in both finite-sample bias and variance.

While economic theory may justify certain restrictions, we take an agnostic approach, letting the data determine the structure. In frequentist univariate settings, clustering methods like k-means (Bonhomme and Manresa, 2015; Hahn and Moon, 2010; Lumsdaine, Okui, and Wang, 2022) effectively impose parameter equality restrictions but are limited to single-dimensional clustering. When parameters span multiple dimensions – such as time and cross-sections in panels or lags and endogenous variables in VARs – co-clustering becomes NP-hard, making standard frequentist methods impractical for high-dimensional economic models where restrictions are inherently multidimensional.

The Bayesian counterpart to clustering can be achieved using Dirichlet Process priors, which can be reinterpreted as sampling equality restrictions alongside parameters. This approach explicitly accounts for uncertainty about which restrictions should hold, allowing the data to simultaneously determine both the clustering structure and the parameter estimates. A key advantage of this framework is its natural extension to multidimensional clustering. To this end, we build on the Matrix Dirichlet Process (MDP) framework of Dunson, Xue, and Carin (2008) proposed for clustering parameters across studies in meta-analysis. We adapt this framework to allow for simultaneous clustering along two dimensions in a VAR setting. We further develop a computationally efficient Gibbs sampler, facilitating the joint estimation of parameter values and restriction structures in high-dimensional models. The MDP prior can easily be combined with other priors, enabling researchers to impose sign restrictions and other identification constraints.

The paper first evaluates the performance of the Matrix Dirichlet Process (MDP) prior for large VAR models using simulations based on three realistic data-generating processes (DGPs): a VAR consistent with the Minnesota prior, and settings with spatial and network-based dependencies. We assess posterior bias, mean squared error (MSE), mean absolute error (MAE), and out-of-sample forecast accuracy measured by root mean squared forecast error (RMSFE), comparing the MDP prior to

both an uninformative prior and the Minnesota prior. The results show that the MDP prior consistently dominates in terms of MSE and MAE across all three DGPs. Even under the Minnesota DGP, the MDP prior outperforms the Minnesota prior due to uncertainty about the shrinkage hyperparameter, and closely approximates the performance of the Oracle-Minnesota prior, which is given knowledge of the true shrinkage parameter. Forecasting results show the MDP prior achieves lower RMSFE than competing methods under both the spatial and network-based DGPs, and outperforms both the Minnesota prior and a FAVAR model for forecasts at longer horizons under the Minnesota-based DGP.

Next, the paper demonstrates the wide applicability of the MDP prior through three empirical applications. The first examines personal income growth across all U.S. states, with a focus on spillover effects and forecasting performance. The uninformative prior yields diffuse posterior distributions, while the Minnesota prior tends to overshrink off-diagonal elements toward zero, often producing implausible estimates of cross-state spillovers. In contrast, the MDP prior uncovers economically intuitive patterns, including stronger spillovers among geographically proximate states – consistent with known regional economic linkages. Given the emphasis on forecasting, we compare RMSFE across priors and find that the MDP prior outperforms the Minnesota prior for all but one state, highlighting its effectiveness in high-dimensional forecasting.

The second application illustrates the method’s usefulness for structural analysis by estimating a medium-scale (18-variable) Bayesian VAR identified using the high-frequency external instrument for monetary policy shocks proposed by Gertler and Karadi (2015). Previous studies have shown that the Minnesota prior performs well for structural analysis in large BVARs when using internal identification schemes, such as recursive orderings (Bańbura, Giannone, and Reichlin, 2010; Crump, Eusepi, Giannone, Qian, and Sbordone, 2021). However, we find that when external instruments – which tend to be noisier – are used, the Minnesota prior shrinks the effects of monetary policy toward zero too aggressively, attenuating the estimated responses. In contrast, the Matrix Dirichlet Process (MDP) prior yields narrow credible intervals while largely preserving the shape and magnitude of the impulse responses relative to those obtained under the uninformative prior. The estimated impulse responses indicate that unexpected monetary policy shocks lead

to declines in inflation, consumer sentiment, and the exchange rate, with effects statistically significant at the 90% level.

The third application examines price spillovers across 24 U.S. industries in response to uncertainty shocks, using monthly industry-level Producer Price Index (PPI) data. This analysis contributes to the literature on sectoral shock propagation (Long and Plosser, 1983; Foerster et al., 2011; Atalay, 2017), which has traditionally relied on structural models to quantify intersectoral linkages. It also relates to studies such as Bloom (2009) and Basu and Bundick (2017), which show empirically that aggregate prices tend to decline in response to uncertainty shocks, typically identified by ordering a volatility index (e.g., the VIX) first in a Cholesky decomposition. However, less is known about how such uncertainty shocks propagate through the production network, which is the focus of this application.

We estimate the model using the MDP prior, an uninformative prior, and the Minnesota prior, and compare posterior impulse response functions. The uninformative prior produces wide credible intervals that hinder inference, while the Minnesota prior imposes strong shrinkage toward zero on off-diagonal elements, as such also shrinking network spillovers to zero. In contrast, the MDP prior leverages clustering to preserve the model’s network structure and yields economically meaningful estimates. We find that prices in upstream sectors respond directly and strongly to uncertainty shocks, while prices in downstream industries and service sectors exhibit weaker responses, primarily driven by spillovers transmitted through the production network.

Related Literature Bayesian priors are widely used in large VARs to manage overparameterization. Bańbura et al. (2010) demonstrate the effectiveness of the Litterman (1979) prior when shrinkage is calibrated to the cross-sectional dimension. Their hyperparameter selection ensures consistent in-sample fit across models. Koop (2013) also finds that the Minnesota prior performs well in forecasting. Chan (2022) refines the Minnesota prior by allowing asymmetric shrinkage, penalizing lags of other variables more than own lags. Giannone, Lenza, and Primiceri (2019) propose a prior specifically to regularize the long-run behavior of the VAR.

Other approaches aim to refine shrinkage priors through selective shrinkage. George, Sun, and Ni (2008) introduce hierarchical priors that stochastically determine which parameters to shrink, building on George and McCulloch (1993). Similarly, Giannone et al. (2021) apply a spike-and-slab prior (though not in a VAR setting), which sets some parameters exactly to zero rather than merely shrinking them. Our paper is also related to Billio, Casarin, and Rossini (2019) who exogenously split the parameters of a VAR model into M blocks (e.g., a block is a lag), and then use Bayesian LASSO for each block, where they use independent Dirichlet Process priors on the Lasso parameters across blocks. The main difference is that the MDP prior identifies similarities between parameters both to impose restrictions and form clusters, moving beyond simple zero restrictions or shrinkage. In addition, unlike many priors, the MDP prior does not assume parameters are i.i.d.; instead, it exploits the matrix structure of these models.

Another way to handle high-dimensional models is through dimensionality reduction techniques. These include dynamic factor models (Bernanke et al., 2005; Forni, Hallin, Lippi, and Reichlin, 2000), principal component analysis (PCA) (Stock and Watson, 2002; De Mol, Giannone, and Reichlin, 2008), and Bayesian compressed regression (Koop, Korobilis, and Pettenuzzo, 2019). However, these methods impose more rigid structures than the clustering framework of the MDP prior, which flexibly groups parameters across equations and variables rather than assuming all variables load on the same set of factors.

Outline In Section 2, we analyze second-order bias in the context of large VAR models, and show how parameter restrictions reduce this bias. Section 3 proposes our MDP priors and discusses a Gibbs Sampling algorithm. In Section 4, we show performance for five realistic data generating processes. Sections 5, 6 and 7 present our applications. Section 8 concludes.

2 Trading off Small Sample Bias with Misspecification Bias

Consider a Vector Autoregression of order one (VAR(1)) with p variables, and let $y_t = (y_{1t}, y_{2t}, \dots, y_{pt})$ be a $p \times 1$ vector observed at time t . We assume a stationary

process

$$y_t = \Phi y_{t-1} + u_t, \quad u_t \sim N(0, \Omega). \quad (1)$$

Rilstone, Srivastava, and Ullah (1996) derive expressions for the second-order bias of estimators that solve moment equations of the form

$$\psi_T(\hat{\beta}) = \frac{1}{T} \sum_{t=1}^T q_t(\hat{\beta}) = 0, \quad (2)$$

with $q_t(\beta)$ a moment function, and ψ_T the sample moment condition. This framework nests the first-order conditions of many maximum likelihood estimators, the orthogonality condition of least squares estimators, and the moment conditions of method of moment estimators, but also can be related to Generalized Method of Moments. To derive the finite-sample moments of such an estimator, Rilstone et al. (1996) find an analytically tractable stochastic approximation for $\hat{\beta}$ of the form

$$\hat{\beta} - \beta_0 = a_{-1/2} + a_{-1} + o_P(T^{-3/2}), \quad (3)$$

where the notation $a_{-s/2}$ denotes a term of order $O_p(T^{-s/2})$ and the second-order bias follows from the expectation of $a_{-1/2} + a_{-1}$. Second-order bias is also sometimes referred to as small-sample bias, as it vanishes as $T \rightarrow \infty$. Expressions for this expansion are derived by implementing a Taylor series expansion of $\psi_T(\hat{\beta}) = 0$ around β_0 and using a Nagar (1959)-type expansion of the inverse of the gradient of the moment function to solve for $\hat{\beta} - \beta_0$ recursively.

Bao and Ullah (2007) extend this framework for time series models where data are not i.i.d., and, among other things, derive expressions for the second-order bias of the VAR(1) model. Assume that $\hat{\beta}$ is $\sqrt{T}$ -consistent and that Assumptions (A)-(C) in Rilstone et al. (1996) hold, which relate to the existence, boundedness and validity of a Lipschitz-condition for the derivatives of q_t around β_0 up to order two and $\psi_T(\beta)$ having an invertible Jacobian that is stochastically bounded around β_0 . Let H_1, H_2 denote the matrices of partial derivatives $H_i = \nabla_i \psi_T$ of order $i = 1, 2$ evaluated at β_0 , with ∇_i the symbol for the partial derivative, and denote $\bar{H}_i = \mathbb{E}(H_i)$. Bao and

Ullah (2007) show (in equation (3) of their paper, using their notation)

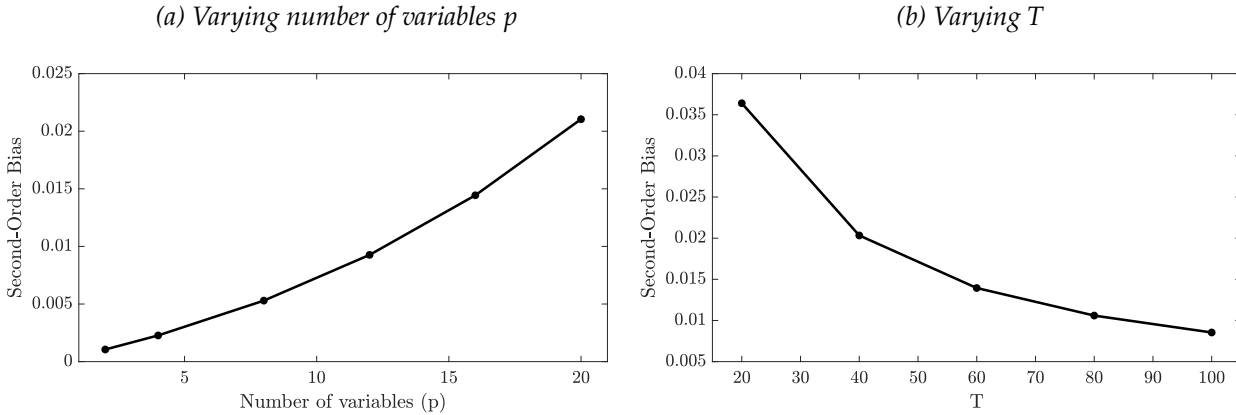
$$a_{-1/2} = -Q\psi_T(\beta_0), \quad \text{with } Q = \bar{H}_1^{-1}, \quad (4)$$

$$a_{-1} = -QV a_{-1/2} - \frac{1}{2}Q\bar{H}_2[a_{-1/2} \otimes a_{-1/2}]. \quad (5)$$

Because $\mathbb{E}(a_{-1/2}) = 0$, the second-order bias of $\hat{\beta}$ is given by $E(a_{-1})$.

Theorem 3 in Bao and Ullah (2007) provides expressions for the second-order bias of a VAR(1) model as in Equation (1). While these expressions are not so easy to interpret, we can use them to compute the theoretically implied second-order bias for specific examples. We visualize how the second-order bias changes, as the data length T and number of variables p vary, in Figure 1. For this particular data-generating process, the bias grows faster than linearly in p ,¹ and decreases at a slower-than-linear rate in T . Hence, not only does estimating a VAR with a large number of variables rapidly increase the variance of our estimators, estimating a VAR with many variables will also result in substantial small-sample bias.

Figure 1: Second-Order Bias of a VAR(1) in element $\hat{\phi}_{12}$



Notes: Figure (a) assumes $T = 50$, Ω has 3 on the diagonal and 1 on the off-diagonal, Φ has 0.7 on the diagonal and 0.3/21 on the off-diagonal. For Figure (b), p is set to 8, Ω is the same and Φ has 0.7 on the diagonal and 0.3/9 on the off-diagonal. y_0 is assumed to be zero.

One possible bias correction approach is to use the expressions for second-order bias to construct a plug-in estimator. This is typically suboptimal from a mean

¹Because adding variables makes the process closer to being non-stationary.

squared error (MSE) efficiency perspective, however, as it increases variance. Bias correction is only desirable when the researcher places a lot of weight on bias reduction relative to variance control, as discussed by Li, Plagborg-Møller, and Wolf (2024) in the context of VARs and Local Projections (Jordà, 2005).

Instead, consider the possibility of imposing a set of linear restrictions $R\beta = r$. In particular, restrictions that two or more coefficients are equal across different variables, e.g. $\phi_{1i} = \phi_{2j}$, assuming that these restrictions are in fact correct. It is well known that such restrictions reduce the variance of the estimator, e.g., one can find a textbook treatment of this result in Hansen (2022). It is also possible to show that such restrictions reduce the second-order bias, as we will do below in Theorem 1. Intuitively, adding linear equality restrictions increases the effective sample size, because we can use more information to estimate the same parameter, hence decreasing the second-order bias. To formalize this, Theorem 1 considers what happens to a_{-1} of the expansion in Equation (3) when imposing a restriction $R\beta = r$ on the estimator. Under the assumption that the restriction holds, this results in a decrease in a_{-1} .

Theorem 1. *Suppose assumptions (A)–(C) of Rilstone et al. (1996) hold, and the true parameter β_0 satisfies the restriction $R\beta_0 = r$. Let a_{-1}^u and a_{-1}^r denote the second-order bias coefficients of the unrestricted and restricted estimators, respectively. Then the difference $\Delta a_{-1} := a_{-1}^u - a_{-1}^r$ can be written as*

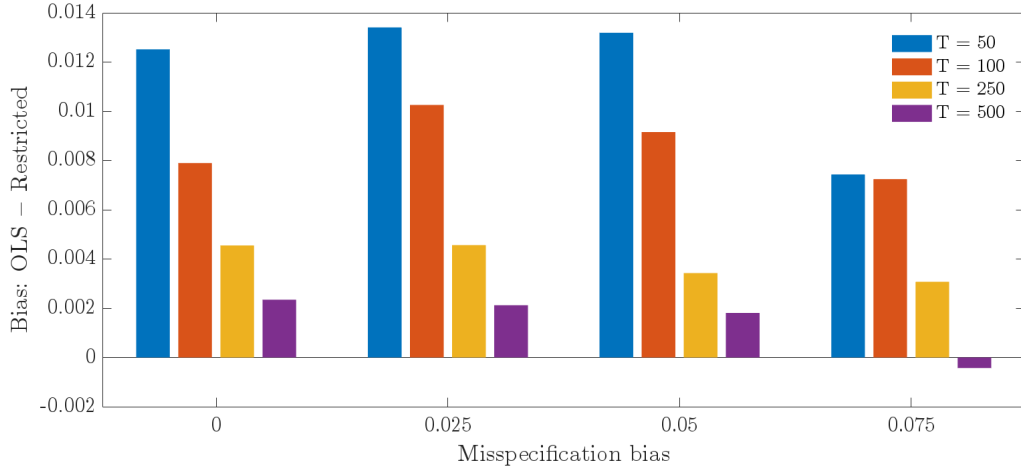
$$\Delta a_{-1} = (Q - Q_r) \left(V a_{-1/2}^u + \frac{1}{2} \bar{H}_2 [a_{-1/2}^u \otimes a_{-1/2}^u] \right),$$

where Q and Q_r are the unrestricted and restricted weighting matrices, and $a_{-1/2}^u$ is the leading bias term for the unrestricted estimator. By positive semi-definiteness of $Q - Q_r$, it follows that imposing the (true) restriction $R\beta = r$ reduces the second-order bias.

Proof: See Appendix A.

Theorem 1 tells us that restrictions of the form $R\beta = r$ can reduce second-order bias if the restrictions hold true. On the other hand, if the restriction is false, a restriction of the form $\phi_{1i} = \phi_{2j}$ introduces misspecification bias proportional to $|\phi_{1i} - \phi_{2j}|$, and, unlike second-order bias, misspecification bias does not vanish

Figure 2: Simulation to illustrate trade-off of second-order sample bias and misspecification bias



Notes: This figure summarizes results from a Monte Carlo simulation (10,000 draws) comparing unrestricted and constrained LS estimators in a stationary VAR(2) model with $p = 3$ variables and different sample sizes T . The true data-generating process (DGP) is specified as:

$$\Phi_1 = \begin{bmatrix} 1 & -0.1 & -0.1 + mbias \\ -0.1 & 0.9 & 0.1 \\ 0.0 & -0.3 & 0.95 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} -0.1 & -0.1 & -0.1 \\ -0.00 & -0.4 & -0.1 \\ -0.1 & 0.1 & -0.2 \end{bmatrix}$$

with innovations $\varepsilon_t \sim \mathcal{N}(0, \Omega)$, where $\Omega = 0.97 \cdot (\mathbf{1} - I) + I$. The restriction imposed in estimation assumes equality of selected off-diagonal elements: $\Phi_1(1, 2) = \Phi_1(1, 3)$ and $\Phi_2(1, 2) = \Phi_2(1, 3)$. When $mbias \neq 0$, these restrictions are incorrect, introducing misspecification bias in the restricted estimator. The x-axis shows different choices for this misspecification bias term. The y-axis shows the resulting estimation bias difference between OLS and constrained least squares, and is computed by first averaging the estimated coefficient matrices across simulations, then subtracting the true parameter matrix. We report the average of the absolute bias across all VAR coefficients.

with the sample size. However, as long as the misspecification bias is smaller than the reduction in second-order bias, introducing restrictions can be bias-reducing in finite samples. We illustrate this point in Figure 2, which displays the average absolute bias across parameters in a three-variable VAR(2) model, as a function of the degree of misspecification in an imposed restriction. The figure demonstrates that, for this data-generating process (DGP), even at moderate sample sizes (e.g., $T = 250$), imposing a misspecified restriction can still reduce overall bias, because the reduction in small-sample bias outweighs the distortion introduced by misspecification.

In practice, the true parameters are unknown, making it difficult to determine the optimal restrictions needed to reduce bias and variance. In addition, when there is

uncertainty about which restrictions to impose, the confidence/credible intervals of the parameters should reflect this. This is where the Matrix Dirichlet Process prior proves useful. The Gibbs Sampling algorithm identifies parameters that are similar and can be clustered. In doing so, the algorithm effectively discovers restrictions that are nearly valid. As outlined above, this *can* lead to improvements in both bias and variance, a result we will demonstrate in our simulations.

3 A Matrix Dirichlet Process Prior for VAR models

Dirichlet Process (DP) priors (Ferguson, 1973) are the Bayesian counterpart of parameter clustering.² Denote the parameters of interest by $\phi_i, i = 1, \dots, N$. In the one-dimensional clustering problem:

$$\phi_i | p \sim \sum_{l=1}^{L_{\max}} p_l \delta_{\bar{\phi}_l}(\phi_i), \quad p \sim DP(\alpha, G_0), \quad \bar{\phi}_l \sim G_0$$

where G_0 is the underlying prior on the parameters ϕ_l , and $\delta_{\bar{\phi}_l}$ is an indicator function, equal to one when ϕ_i is in cluster l , and zero else. Parameters in the same cluster ($l = 1, \dots, L_{\max}$) take on the same value, which we denote by $\bar{\phi}_l$. $L_{\max}$ is a truncation, as the untruncated DP prior has $L_{\max} \rightarrow \infty$.³ The probability of being assigned to cluster l is denoted by p_l ,

$$p_l = \beta_l \prod_{k=1}^{l-1} (1 - \beta_k), \quad l = 1, \dots, L_{\max}, \quad \beta_l \sim \text{Beta}(1, \alpha)$$

A lower value of α results in fewer expected clusters, meaning that more parameters are likely to be shared. This can be intuitively understood through the stick-breaking representation of the Dirichlet Process (DP) prior. In this framework, the probability of an observation being assigned to cluster l , denoted as p_l , is determined by β_l , which represents the length of the l th piece in a sequential stick-breaking process. When α is small, the initial stick segments tend to be longer,

²Not to be confused with clustering standard errors, which is a method used to adjust standard errors in the presence of dependence among grouped observations.

³Careful choice of $L_{\max}$ and hyperprior probabilities ensures that this truncation approximation is sufficiently accurate.

leading to fewer total breaks. Consequently, more observations are assigned to the same cluster, reinforcing the tendency toward fewer distinct clusters.

We wish to apply this prior to a VAR model, that is, our parameters of interest now come from a matrix Φ with elements ϕ_{ij} . The DP prior can be applied directly to matrices *if* we want to assume all elements are independent. However, when we want to take the matrix structure of the model into account, Matrix Dirichlet Process priors are a more appropriate choice.

We build on Dunson et al. (2008), who propose a Matrix Dirichlet Process Prior in the context of meta-analysis. A key distinction between their framework and ours lies in the clustering structure. In meta-analysis, their approach allows clustering across studies (equations) but not within them, meaning that while some parameters may be shared across studies, there is no reason within their framework why one would want to cluster parameters of different explanatory variables within or across studies. In contrast, in the VAR setting, we aim to allow clustering in both the row and column dimensions of the parameter matrix, enabling a more flexible dependence structure that is appropriate for VAR models.

We follow the notation in Dunson et al. (2008) closely. For simplicity, consider that we only have one lag in our VAR model, with parameter matrix Φ . Our Matrix Dirichlet Process (MDP) prior is given by:

$$\phi_{ij}|G \sim \sum_{l=1}^{L_{\max}} G_{ijl} \delta_{\bar{\phi}_l}(\phi_{ij}), \quad G \sim \text{MDP}(\alpha, \gamma, G_0), \quad \bar{\phi}_l \sim G_0$$

where the probability that ϕ_{ij} is in cluster l is given by

$$G_{ijl} = \beta_{ijl} \left\{ \prod_{l < h} (1 - \beta_{ijh}) \right\} \quad (6)$$

with $\beta_{ijh} = \alpha_{ih} \gamma_{jh}$, where $\alpha_{ih} \sim \text{Beta}(1, \alpha)$ and $\gamma_{jh} \sim \text{Beta}(1, \gamma)$. The parameters α_{ih} and γ_{jh} measure the tendency to allocate i to cluster h and the tendency to allocate j to cluster h , respectively. The MDP prior is in principle generalizable to n dimensions, with β equal to the product of n parameters drawn from n separate Beta distributions.

As in the univariate Dirichlet Process prior, the equation for G_{ijl} follows the stick-breaking process, where lower values of α and γ lead to fewer expected clusters. While α_{ih} and γ_{jh} are independent, the variables β_{ijh} are not, as they determine the prior probabilities of parameter clustering.

One attractive feature of this prior framework is its flexibility in allowing asymmetry between prior and posterior clustering strength. Specifically, one can set $\alpha = \gamma$ or allow them to differ. In a VAR setting, this means controlling whether clustering is stronger across rows or columns of the coefficient matrix. Setting $\alpha < \gamma$ implies the researcher expects greater similarity across equations than of parameters within individual equations. As a researcher may want to remain agnostic about the prior clustering strength, one can also put priors on the distributional parameters α and γ themselves, and sample them along with the other parameters and cluster assignments.

As in Dunson et al. (2008), the Matrix Dirichlet Process prior has the following prior probabilities for two parameters ϕ_{ik} and ϕ_{jk} being clustered together

$$P(\phi_{ik} \in l \text{ and } \phi_{jk} \in l) = \frac{1}{(\alpha + 1)(\gamma + 2) - 1}$$

and has the desirable property that

$$P(\phi_{ik} \in l \text{ and } \phi_{jk} \in l | \phi_{im} \in q \text{ and } \phi_{jm} \in q) > P(\phi_{ik} \in l \text{ and } \phi_{jk} \in l),$$

that is, if i and j are similar in one dimension/attribute (k), it is also more likely they are similar in another dimension/attribute (m). This is particularly attractive in the context of VARs, because parameters are likely dependent; if variable i and j are similarly exposed to variable k , then they are also more likely similarly exposed to variable l . This dependence can be captured with the MDP prior.

Another advantage of this approach is that it is not necessary to abandon other priors that one might want to use – for example, for the identification of shocks – in order to reap the benefits from clustering. The MDP prior can be combined with other popular priors, like the Minnesota prior, or with priors that impose sign restrictions, by careful choice of G_0 .

For ease of interpretation, the notation above focuses on the case where clustering is performed across two dimensions with a single lag (VAR(1)). However, various approaches can be used to extend this framework to multiple lags, such as in a VAR(k). Our approach clusters over pairs (i, j) rather than individual parameters, meaning that each cluster represents a vector of coefficients across the k different lags. Alternatively, one could apply independent MDP priors to each lag separately. A third approach introduces a dependency structure in the MDP prior, ensuring that variable pairs (i, j) clustered together at one lag are also more likely to be clustered together at other lags. We favor the first approach due to its parsimony.

3.1 A Computationally Efficient Gibbs Sampler

When estimating large Bayesian VARs, it is important to have an algorithm that is computationally efficient. One can imagine two-dimensional clustering can become computationally burdensome, and, in fact, infeasible. For our version of the MDP prior and VAR models we adapt the sampling algorithm in Dunson et al. (2008), built on Ishwaran and James (2001), in several dimensions, to allow for a fast Gibbs sampling algorithm.

Jointly drawing coefficients and clusters is challenging, but the iterative process simplifies this by leveraging conditional posteriors. Given a known form for the conditional posterior, the algorithm first draws cluster assignments conditional on the parameters, then samples parameters conditional on the clusters. Iterating avoids the need for acceptance/rejection criteria, unlike approximate samplers such as random walk Metropolis-Hastings, which rely on such schemes and can have slow convergence. The closed-form conditional posteriors and Gibbs sampling algorithm make this approach tractable and computationally efficient.

A VAR(k) model for p variables is:

$$y_t = \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \cdots + \Phi_k y_{t-k} + u_t, \quad t = 1, \dots, T \quad (7)$$

with $u_t \sim \mathcal{N}(0, \Omega)$. We can stack our Vector Autoregression as: $Y = XB + \varepsilon$, with

$$y = \begin{bmatrix} y'_k \\ y'_{k+1} \\ \vdots \\ y'_T \end{bmatrix}, \quad X = \begin{bmatrix} y'_{k-1} & y'_{k-2} & \cdots & y'_1 \\ y'_k & y'_{k-1} & \cdots & y'_2 \\ \vdots & \vdots & \ddots & \vdots \\ y'_{T-1} & y'_{T-2} & \cdots & y'_{T-k} \end{bmatrix}, \text{ and } B = \begin{bmatrix} \Phi'_1 \\ \Phi'_2 \\ \vdots \\ \Phi'_k \end{bmatrix},$$

where

$$y'_t = \begin{bmatrix} y_{1,t} & y_{2,t} & \cdots & y_{p,t} \end{bmatrix}.$$

and $\varepsilon \sim N(0, \Sigma)$ where Σ is a block diagonal matrix with blocks Ω .

For the regression parameters B , we use the conjugate normal prior and for Σ ,

$$p(\Sigma) \propto |\Sigma|^{-(p+1)/2}.$$

This prior for Σ is the matrix generalization of the prior $p(\sigma^2) \propto \sigma^{-2}$, commonly used in univariate settings.

For the Gibbs sampler, we introduce three sets of additional latent variables that simplify the sampling algorithm. Denote $C_{ij} = h$ as the cluster assignment indicator, taking values $h \in \{1, \dots, L_{\max}\}$, indicating that variable pair (i, j) has been assigned to cluster number h . We have

$$C_{jm} = \min\{l : S_{jml} = T_{jml} = 1\},$$

with $S_{jml} \sim \text{i.i.d. Bernoulli}(\alpha_{ml})$ and $T_{jml} \sim \text{i.i.d. Bernoulli}(\gamma_{jl})$, for $l = 1, \dots, L_{\max} - 1$, and $S_{jmL_{\max}} = T_{jmL_{\max}} = 1$.

The Gibbs sampler iterates between the following four steps:

- (1) Sample latent variables $S_{jml}, T_{jml}, \alpha_{mh}, \gamma_{jh}$ and the cluster assignments C_{jm} conditional on the previous draw of Σ, B , for all $j, m = 1, \dots, p, j \neq m$.
- (2) Sample hyperparameters α and γ conditional on cluster assignments $\{C_{ij}\}$.

Note that this step is optional, and only needed if one uses a hyperprior on α and γ . Sampling α and γ is standard if one uses the conjugate Gamma prior (West, 1992).

Throughout all simulations, as well as our first application, we use a hyperprior $\text{Gamma}(1,1)$ for both α and γ , which is fairly diffuse and lets the data speak on how much to cluster. As in West (1992), after having sampled α_{mh} and γ_{jh} in the previous step, we can update the posteriors and draw a new α and γ .

- (3) Sample B conditional on the last draw of Σ and the cluster assignments C_{ij} .
- (4) Sample Σ conditional on the last draw of B .

This is standard under the prior we assume; the full conditional posterior density of Σ is given by:

$$p(\Sigma|B, Y) \propto |\Sigma|^{-(T-k+p+1)/2} \exp\left(-\frac{1}{2} \text{tr} [\Sigma^{-1}(Y - XB)'(Y - XB)]\right),$$

hence, we sample Σ from an inverted Wishart distribution with parameters:

$$S = (Y - XB)'(Y - XB) \quad \text{and} \quad T - k \text{ degrees of freedom.}$$

Next, we elaborate on steps (1) and (3) in detail.

3.2 Step (1): Sampling cluster assignments C_{jm}

We sample S_{jmh} and T_{jmh} by setting them to 1 for $h = C_{jm}$, and sampling them with probabilities $\kappa_{st} = P(S_{jmh} = s, T_{jmh} = t)$ for $h = 1, \dots, C_{jm} - 1$, where

$$\begin{aligned} \kappa_{00} &= \frac{(1 - \alpha_{jh})(1 - \gamma_{mh})}{(1 - \alpha_{jh}\gamma_{mh})} \\ \kappa_{10} &= \frac{\alpha_{jh}(1 - \gamma_{mh})}{(1 - \alpha_{jh}\gamma_{mh})} \\ \kappa_{01} &= \frac{(1 - \alpha_{jh})\gamma_{mh}}{(1 - \alpha_{jh}\gamma_{mh})}, \end{aligned}$$

and $\kappa_{11} = 1 - \kappa_{00} - \kappa_{10} - \kappa_{01}$. After sampling S_{jmh} and T_{jmh} , we can sample α_{jh} and γ_{mh} from their conditional posteriors:

$$\begin{aligned}\alpha_{jh} | \dots &\sim \text{Beta} \left(1 + \sum_{m: C_{jm} \geq h} S_{jmh}, \alpha + \sum_{m: C_{jm} \geq h} (1 - S_{jmh}) \right) \\ \gamma_{mh} | \dots &\sim \text{Beta} \left(1 + \sum_{m: C_{jm} \geq h} T_{jmh}, \gamma + \sum_{m: C_{jm} \geq h} (1 - T_{jmh}) \right)\end{aligned}$$

With the sampled α_{jh} 's and γ_{mh} 's in hand, one can compute $\beta_{jmh} = \alpha_{jh}\gamma_{mh}$, and G_{jml} follows from Equation (6). At last, cluster assignments C_{jm} follow a categorical distribution, with the conditional probability that pair (j, m) is assigned to cluster h given by:

$$P(C_{jm} = h | \cdot) = \frac{G_{jmh} \pi(Y | X, B_{(jm)} = B_{(h)}, \cdot)}{\sum_{k=1}^{L_{\max}} G_{jmk} \pi(Y | X, B_{(jm)} = B_{(k)}, \cdot)}, \quad (8)$$

where $\pi(\cdot)$ denotes the posterior of the VAR model, evaluated when the coefficient block $B_{(jm)}$ for pair (j, m) is set to the value associated with cluster h , and all other parameters remain fixed at their current values. This element-wise evaluation is repeated for each possible cluster $h = 1, \dots, L_{\max}$, and becomes computationally expensive when both the number of pairs (j, m) and $L_{\max}$ are large, which is the case in our setting. In the next subsection, we discuss a computationally tractable solution.

3.2.1 Multiple-Try Metropolis

The computational bottleneck in generating draws for cluster assignments C_{jm} from the conditional posterior in Equation (8) comes from the need to evaluate the likelihood under all possible cluster assignments $L_{\max}$. A more computationally efficient alternative is to use a Markov Chain Monte Carlo sampler that only requires evaluating $\pi(Y | X, B_{(jm)} = B_{(h)}, \cdot)$ for a small subset of candidate clusters $S_{\text{sub}}^{(jm)} \subset \{1, \dots, L_{\max}\}$ at each update.

Assume that in the previous iteration t , pair (jm) was assigned to cluster $h^{(t)}$, i.e., $C_{jm}^{(t)} = h^{(t)}$. Consider the extreme setting where the subset S_{sub} consists of a single proposal candidate $h^* \sim q(h)$, reducing the procedure to a standard Metropolis–Hastings (MH) step. Note that while the framework allows for dependence of the proposal distribution on the current draw, i.e., $q(h|h^{(t)})$, in what follows we focus on what is called an *independence sampler*, that is, $q(h|h^{(t)}) = q(h)$. A proposed cluster h^* is drawn from a proposal distribution $q(h)$ (e.g., uniform $q(h) \propto 1$), and the acceptance probability is

$$\alpha = \min \left(1, \frac{G_{jm h^*} \pi(Y | X, B_{(jm)} = B_{(h^*)}, \cdot)}{G_{jm h^{(t)}} \pi(Y | X, B_{(jm)} = B_{(h^{(t)})}, \cdot)} \times \frac{q(h^{(t)})}{q(h^*)} \right). \quad (9)$$

Note that the proposal ratio $q(h^{(t)})/q(h^*)$ cancels if $q(h)$ is uniform over $\{1, \dots, L_{\text{max}}\}$. This method requires only two likelihood evaluations per pair (jm) (for the proposed and current draw), but mixes poorly and converges slowly when L_{max} is large, as it explores only one alternative option per sampler iteration.

Intuitively, it would be desirable instead to propose a subset $S_{\text{sub}}^{(jm)}$ of size $M > 1$, with $M \ll L_{\text{max}}$, and adjust the sampler accordingly. This idea fits naturally within the framework of the Multiple-Try Metropolis (MTM) algorithm (Liu, Liang, and Wong, 2000), adapted here to the setting of generating draws from a categorical variable with a large number of categories.

More concretely, at each update, a subset $S_{\text{sub}}^{(jm)} = \{h_1, \dots, h_M\}$ of candidate clusters is proposed, where for now we assume each candidate $h_i \sim q(h)$ is drawn independently with replacement. Each candidate is assigned an unnormalized importance weight

$$w_i = \frac{G_{jm h_i} \cdot \pi(Y | X, B_{(jm)} = B_{(h_i)}, \cdot)}{q(h_i)},$$

and a proposed cluster $h^* \in S_{\text{sub}}^{(jm)}$ is selected with probability proportional to w_i . To compute the acceptance probability, a reverse set $S_{\text{rev}} = \{h'_1, \dots, h'_M\}$ has to be generated as: $h'_1 = h^{(t)}$ (the current state) and $h'_2, \dots, h'_M \sim q(h)$ independently, with replacement. Reverse weights w'_i are computed in the same way as above.

The acceptance probability for proposal draw h^* then becomes:

$$\alpha = \min \left(1, \frac{\sum_{i=1}^M w_i}{\sum_{i=1}^M w'_i} \right). \quad (10)$$

Note that for discrete random variables, this sampler has the attractive property that if we draw the proposals for $S_{\text{sub}}^{(jm)}$ *without* replacement, then as the subset size $M \rightarrow L_{\text{max}}$, the acceptance rate $\alpha \rightarrow 1$, since in the limit we recover the posterior of Equation (8).⁴

One concern arises in settings where both the number of pairs (jm) , with $j = 1, \dots, p$, $m = 1, \dots, p$, $j \neq m$, and the number of possible clusters L_{max} are large. When the proposal subset size M is relatively small, the probability that two distinct pairs (jm) and (kl) draw overlapping subsets, i.e., $S_{\text{sub}}^{(jm)} \cap S_{\text{sub}}^{(kl)} \neq \emptyset$, becomes small under uniform or otherwise uninformative proposal distributions. As a result, it becomes difficult for the sampler to assign multiple pairs to the same cluster, and hence detect groups of similar variables.

A practical solution to this problem is to use the prior G_{jmh} as the proposal distribution, i.e., $q(h) \propto G_{jmh}$. Since the Matrix Dirichlet Process prior encourages clustering across pairs, it will *oversample* clusters in a way that increases the probability that multiple pairs share overlapping clusters, i.e., $S_{\text{sub}}^{(jm)} \cap S_{\text{sub}}^{(kl)} \neq \emptyset$ even when $M \ll L_{\text{max}}$. The acceptance probability will correct for this oversampling through the importance weights. In particular, if we choose the proposal distribution to be the prior, i.e., $q(h) \propto G_{jmh}$, then each weight simplifies to

$$w_i = \pi(Y \mid X, B_{(jm)} = B_{(h_m)}, \cdot),$$

⁴In continuous state spaces, a high acceptance rate is often associated with poor exploration around the current draw. However, in our setting, we use an independence sampler, and, furthermore, the state space is discrete and finite. In this case, a high acceptance rate indicates that the proposal set S_{sub} captures regions of high posterior mass well. As the subset better approximates the conditional posterior, the acceptance rate increases, reflecting effective updates.

i.e., the likelihood, such that the acceptance probability reduces to

$$\alpha = \min \left(1, \frac{\sum_{i=1}^M \pi(Y | X, B_{(jm)} = B_{(h_i)}, \cdot)}{\sum_{i=1}^M \pi(Y | X, B_{(jm)} = B_{(h'_i)}, \cdot)} \right). \quad (11)$$

That is, the acceptance ratio depends solely on the likelihood terms. This simplification arises because the prior is already reflected in how the subset of proposal draws is generated; the MH correction now only needs to adjust for differences in the likelihood.

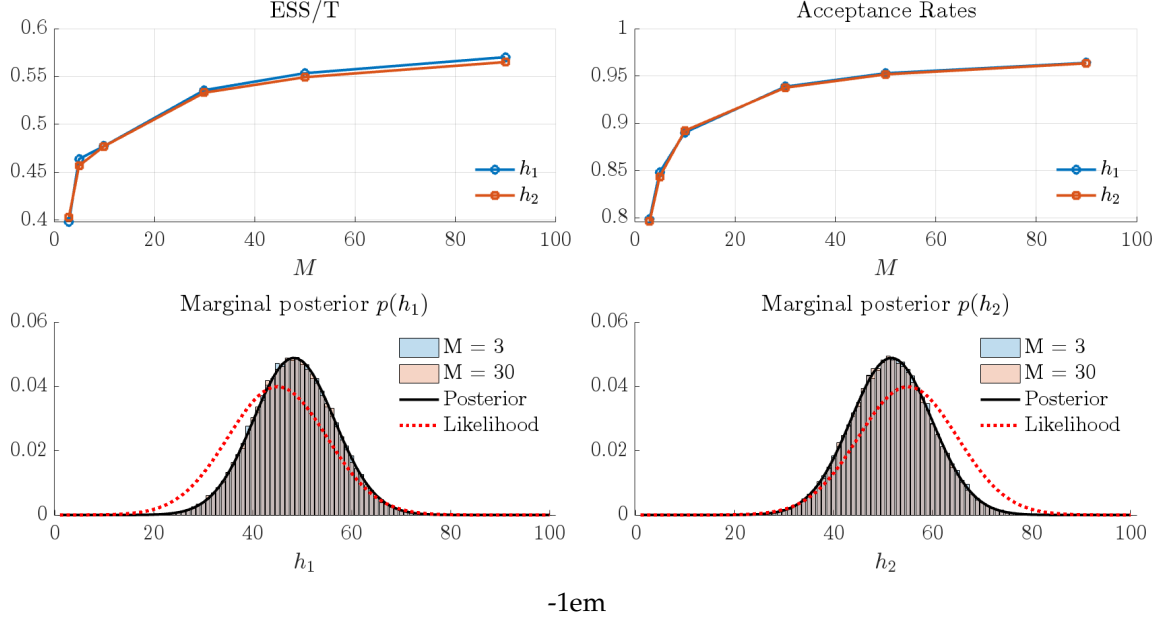
Figure 3 shows simulation results from a toy example where a Gibbs sampler with Multiple-Try Metropolis-Hastings is applied to sample from a distribution over two discrete latent variables taking values in $\{1, \dots, 100\}$. The likelihood favors values of h_1 and h_2 near their fixed means μ_1 and μ_2 , while the prior encourages the two variables to be close. We vary the number of proposal draws M in the Multiple-Try algorithm used, and examine its effect on the effective sample size (ESS – a measure of sampler quality that penalizes high autocorrelation in the samples), acceptance rates, and posterior estimates. Proposals are drawn from the prior. Even for small values of M , such as $M = 3$, the estimated marginal posteriors closely match the true distributions; the acceptance rate remains high (around 80%), and the ESS is only about 30% lower than when using as many as 90 proposal draws instead. Since each step requires only $2M$ likelihood evaluations, $M = 3$ versus $M = 90$ results in a 30-times reduction in function evaluations per iteration.

3.3 Step (3): Sampling B conditional on cluster assignment and previous Σ

First, assume there are no clusters, and assume a conjugate prior for B . In that case, the Gibbs sampler for the VAR is completely standard, and we sample from a matricvariate normal distribution. More specifically, if we use a conjugate prior $B | \Sigma \sim MN(BP, \Sigma \otimes Q)$, we need to sample from the matricvariate normal distribution with mean $(X'X + Q^{-1})^{-1}(X'Y + Q^{-1}BP)$ and covariance $\Sigma \otimes (X'X + Q^{-1})^{-1}$. Write

$$B \sim MN(M, \Sigma \otimes V)$$

Figure 3: Multiple-Try Metropolis for sampling from a categorical posterior.



Notes: We apply a Gibbs sampler with Multiple-Try Metropolis-Hastings (MTMH) to discrete variables $h_1, h_2 \in \{1, \dots, 100\}$, varying M which is the number of proposal draws. The joint prior and likelihood are respectively:

$$p(h_1, h_2) \propto \exp\left(-\frac{(h_1 - h_2)^2}{2\tau^2}\right), \quad \ell(h_1, h_2) \propto \exp\left(-\frac{(h_1 - \mu_1)^2}{2\sigma^2}\right) \times \exp\left(-\frac{(h_2 - \mu_2)^2}{2\sigma^2}\right),$$

with $\tau = 10$, $\mu_1 = 45$, $\mu_2 = 55$, and $\sigma = 10$. Effective sample size (ESS) is computed using the standard formula

$$ESS = \frac{T}{1 + 2 \sum_k \rho_k},$$

where ρ_k is the lag- k autocorrelation of the centered trace, truncated at the first negative value.

where:

$$M = (X'X + Q^{-1})^{-1}(X'Y + Q^{-1}BP),$$

$$V = (X'X + Q^{-1})^{-1}.$$

Next, we introduce the cluster assignments, drawn in step (1) of the algorithm as C_{ij} 's. These cluster assignments can be written as a set of linear equality constraints: $RB = r$. This representation is convenient, because we can use the following. The conditional posterior of B is still matricvariate normal, but now with mean $M + VR'(RVR')^{-1}(r - RM)$ and covariance $(\Sigma \otimes V) - (\Sigma \otimes VR')(RVR')^{-1}(\Sigma \otimes RV)$.

3.4 Posterior consistency

Hjort, Holmes, Müller, and Walker (2010) discuss how Schwartz’s (1965) theorem can be applied to establish posterior consistency for Dirichlet Process priors. In this context, the posterior probability that two well-separated parameters, say Φ_{ij} and Φ_{kl} with $d(\Phi_{ij}, \Phi_{kl}) > \varepsilon_T$, are incorrectly clustered together converges to zero at an exponential rate, provided two conditions hold:

1. **Prior Positivity:** The prior assigns positive probability to the configuration in which the parameters are not clustered together.
2. **Testing Condition:** There exist uniformly exponentially consistent tests for distinguishing between Φ_{ij} and Φ_{kl} .

Ghosal and van der Vaart (2007) extend Schwartz’s theorem to allow for non-i.i.d. data. Under these assumptions, a stable VAR with Gaussian errors and a suitable hyperprior on the clustering parameters satisfies the conditions above. It follows that if there is no true clustering structure and the parameters are well separated, the posterior probability of assigning them to the same cluster converges to zero exponentially fast as the sample size increases.

4 Performance Evaluation on Simulated Data

To assess the performance of the MDP prior and compare it to existing methods, we construct three realistic data-generating processes (DGPs) for VAR(1) and VAR(2) models. We use $N = 15, T = 50$, for which we generate 100 synthetic datasets for each DGP. Across these experiments, we systematically vary Φ , the coefficient matrix of the VAR, and Ω , the variance-covariance matrix of the innovations. The choice of moderate N combined with small T allows the models to mimic the estimation challenges typically encountered in large VARs.

For each synthetic dataset, we simulate data of length $T + T_{\text{out}}$ based on the corresponding VAR model. We evaluate both the bias of the posterior mean and the variance of the posterior estimates, alongside the forecasting performance on $T_{\text{out}} = 100$ out-of-sample forecasts. We use 10,000 draws for each sampler, discarding the first 2,000 as burn-in.

We compare our Bayesian clustering framework with two alternatives: (i) an uninformative prior and (ii) the conjugate variation of the Minnesota prior. We follow the approach in Bańbura et al. (2010) to select the hyperparameter of the Minnesota prior which governs the tightness of the own and cross-lags. For the prior mean of the diagonal elements, we provide the Minnesota prior with the *correct* DGP-implied mean, reflecting that researchers often have a reasonable prior guess for this parameter. We do the same for the Matrix Dirichlet Process prior on the unrestricted diagonal elements. The prior variance for the MDP diagonal and off-diagonal elements is set to 1, which is wide for these specific DGPs, and ensures the MDP prior is clustering rather than shrinking. For forecasting, we also compare the performance to a FAVAR model, where we generate 1, 2, 3 or 4 factors using principal component analysis, and regress the outcome variables on lags of those factors. The results present the best out of the four options.

The first DGP is based on the Minnesota prior, which we use to draw Φ (the coefficient matrix) and Ω (the variance-covariance matrix) for the synthetic datasets.

- (i) **DGP 1: Minnesota VAR** – Φ and Ω are drawn from a Minnesota prior with $\lambda = 0.1$, diagonal mean 0.5, and a decay parameter of 2, corresponding to a standard specification of the Minnesota prior.

The motivation for considering a data-generating process (DGP) consistent with the structure of a Minnesota prior stems from the expectation that a Bayesian framework employing this prior would yield the most accurate inference. However, as shown in the results below for DGP 1, the Matrix Dirichlet Process (MDP) prior outperforms the Minnesota prior in several of the metrics we consider, once uncertainty around the regularization hyperparameter is taken into account. For completeness, we also compare the MDP prior to the *Oracle-Minnesota* prior, which is given the true regularization hyperparameter used in the simulation.

The remaining two DGPs are based on spatial and network structures, motivated by the empirical applications presented below. For both, we choose Ω to have 3's on the diagonal and 1's as off-diagonal elements, and for the coefficients Φ :

- (ii) **DGP 2: Spatial VAR** – This DGP represents a spatial VAR model in which the entries of Φ reflect decayed influence based on the spatial proximity of variables. For each pair of variables i and j , the coefficients are determined as:

$$\phi_{ij} = c \cdot \exp(-b \cdot |i - j|),$$

where $b = 0.3$ controls the decay rate, $c = 0.5$ scales the overall influence, and $|i - j|$ is the absolute difference between indices i and j . This structure captures the notion that variables closer in index exert a stronger influence on one another, while the influence decays exponentially with distance. For the VAR(2), we assume the same structure for the first lag, and for the second lag, we multiply c by 0.05.

- (iii) **DGP 3: Network VAR** – This DGP is informed by our third empirical application and is based on a network VAR structure, where the coefficient matrix Φ is constructed from the 15-sector Input-Output Table published by the Bureau of Economic Analysis (2023). The adjacency matrix derived from this dataset is scaled by total flows to normalize the influence across nodes. To ensure stationarity, the resulting matrix is further multiplied by $\alpha=0.9$ for 15 sectors for a VAR(1), or $\alpha = [0.9, 0.05]$ for a VAR(2). This setup is inspired by the network VAR (NVAR) framework in Mlikota (2023), which models spillovers between variables as proportional to the strength of their connections in the network. In particular, as in Mlikota (2023):

$$y_t = \alpha_1 A y_{t-1} + \dots + \alpha_p A y_{t-p} + u_t$$

with A an observed or estimated network and $\alpha_i, i = 1, \dots, p$ a scalar.

Intuitively, we expect the MDP prior to be better suited for data structures such as these, as it accounts for dependencies among parameters – an aspect the Minnesota prior neglects by assuming parameter independence. While alternative priors specifically tailored to the respective DGPs may also perform well, a key appeal of the MDP prior is its consistent performance across a diverse range of DGPs including when the assumed DGP is misspecified.

Table 1 reports the bias of the posterior means for the diagonal and off-diagonal elements of the coefficient matrix under the uninformative prior, Minnesota prior, and MDP prior across all three DGPs and two lag lengths ($p = 1, 2$). This separation highlights how each method performs under different dependency structures. The table also presents the mean squared and absolute error (MSE and MAE), computed across all posterior draws relative to the true parameter values.

The results in Table 1 reveal several consistent patterns when comparing the Matrix Dirichlet Process (MDP) prior to the Minnesota prior with unknown hyperparameter. First, the MDP outperforms the Minnesota prior across all settings in terms of mean squared error (MSE) and mean absolute error (MAE) of the posterior draws for the coefficient matrix. In particular, under DGP 1, the MDP’s performance is consistently closer to that of the Oracle-Minnesota prior, which is given the true regularization hyperparameter. Second, the MDP prior exhibits uniformly lower absolute bias on the off-diagonal elements of the coefficient matrix. Recall that both the diagonals of the Minnesota prior and MDP prior are centered around the true average diagonal. The Minnesota prior tends to yield lower bias on the diagonal elements, and both MDP and Minnesota consistently improve upon the uninformative prior. These findings align with the theoretical results presented in Section 2, which show that even (mildly) misspecified restrictions can reduce overall bias when small-sample bias is sufficiently large.

Next, we evaluate out-of-sample forecast performance at horizons $h = 1, 2, 4$ using the Root Mean Squared Forecast Error (RMSFE), averaged across all variables i and across all posterior draws. Table 2 summarizes the results. For the empirically motivated DGPs 2 and 3, the Matrix Dirichlet Process (MDP) prior consistently yields lower RMSFE on average than the Minnesota prior, the uninformative prior, and the FAVAR model across all forecast horizons. Under the Minnesota DGP, the FAVAR performs well at the one-step-ahead horizon ($h = 1$), while the MDP prior achieves superior forecast accuracy at longer horizons.

Table 1: Comparing in-sample performance – simulations for three DGPs, two lag structures, and different prior choice. Lower values indicate better performance. O-Minnesota is when we give the Minnesota-prior the correct shrinkage parameter (oracle, only applicable for the Minnesota DGP).

VAR(p)	Statistic	Uninformative	O-Minn.	Minnesota	MDP
Minnesota ($N = 15, T = 50$)					
$p = 1$	MSE	0.135	0.012	0.030	0.014
	MAE	0.277	0.085	0.131	0.088
	Abs. Bias (Diag)	0.201	0.055	0.093	0.109
	Abs. Bias (Off-diag)	0.174	0.055	0.086	0.081
$p = 2$	MSE	0.709	0.009	0.028	0.010
	MAE	0.610	0.071	0.127	0.071
	Abs. Bias (Diag)	0.308	0.047	0.083	0.105
	Abs. Bias (Off-diag)	0.283	0.047	0.076	0.061
Spatial ($N = 15, T = 50$)					
$p = 1$	MSE	0.113	—	0.028	0.009
	MAE	0.264	—	0.125	0.071
	Abs. Bias (Diag)	0.163	—	0.087	0.140
	Abs. Bias (Off-diag)	0.162	—	0.089	0.059
$p = 2$	MSE	0.431	—	0.036	0.003
	MAE	0.494	—	0.143	0.031
	Abs. Bias (Diag)	0.224	—	0.104	0.126
	Abs. Bias (Off-diag)	0.223	—	0.105	0.022
Network ($N = 15, T = 50$)					
$p = 1$	MSE	0.106	—	0.026	0.012
	MAE	0.255	—	0.116	0.075
	Abs. Bias (Diag)	0.163	—	0.126	0.129
	Abs. Bias (Off-diag)	0.156	—	0.082	0.061
$p = 2$	MSE	0.439	—	0.030	0.008
	MAE	0.497	—	0.127	0.055
	Abs. Bias (Diag)	0.231	—	0.102	0.128
	Abs. Bias (Off-diag)	0.226	—	0.090	0.043

Table 2: Comparing forecasting performance – RMSFE at forecast horizons $h = 1, 2, 4$, for 100 out-of-sample forecasts. Simulation for three DGP's, two lag structures, and different prior choice. Lower values indicate better forecast accuracy. Oracle-Minnesota is when we give the Minnesota-prior the true shrinkage parameter (only applicable for the Minnesota DGP).

VAR(p)	Horizon	Uninformative	O-Minn.	Minnesota	FAVAR	MDP
Minnesota (N=15, T=50)						
$p = 1$	$h = 1$	0.572	0.422	0.450	0.436	0.456
	$h = 2$	0.689	0.472	0.511	0.524	0.506
	$h = 4$	0.883	0.528	0.585	0.643	0.550
$p = 2$	$h = 1$	1.179	0.457	0.514	0.467	0.504
	$h = 2$	2.094	0.510	0.586	0.557	0.564
	$h = 4$	13.29	0.574	0.689	0.701	0.616
Spatial (N=15, T=50)						
$p = 1$	$h = 1$	2.603	—	1.978	1.971	1.742
	$h = 2$	2.991	—	2.173	2.372	1.932
	$h = 4$	3.774	—	2.528	3.009	2.258
$p = 2$	$h = 1$	6.570	—	3.378	3.474	2.899
	$h = 2$	11.17	—	3.621	4.833	3.083
	$h = 4$	79.77	—	4.224	8.344	3.506
Network (N=15, T=50)						
$p = 1$	$h = 1$	2.651	—	1.977	1.989	1.790
	$h = 2$	3.048	—	2.150	2.335	1.959
	$h = 4$	3.713	—	2.372	2.789	2.192
$p = 2$	$h = 1$	5.684	—	2.311	2.195	1.874
	$h = 2$	10.13	—	2.579	2.691	2.122
	$h = 4$	76.15	—	2.944	3.397	2.466

5 First application: Forecasting State-Level Personal Income

In the first application, quarterly state-level personal income data for the United States from 1948 to 2023 are analyzed.⁵ Accurate forecasts of state-level income are essential for policy planning, including tax revenue projections and the allocation of federal funds. The analysis also provides an opportunity to empirically assess two commonly posited forms of spatial and economic connectivity: whether geographically proximate states are more closely linked, and whether economically larger states tend to propagate shocks more broadly through the national economy than smaller states. The MDP prior is particularly well-suited to this context, as it explicitly models cross-variable dependencies. In particular, the MDP prior captures the logic that if two states, i and j , exhibit similar responses to a shock in state k , then they are also likely to respond similarly to a shock in another state m .

A Bayesian Vector Autoregression is estimated for the demeaned growth rates of personal income to model cross-state dependencies, using three alternative prior specifications: the Matrix Dirichlet Process prior, an uninformative prior, and the Minnesota prior, the latter again tuned as in Bańbura et al. (2010). The resulting models are high-dimensional, with 51 variables (one per state, plus the District of Columbia). The VAR is estimated with two lags, because the data are quarterly and growth rates have little persistence. Because the model is specified in growth rates, the Minnesota prior shrinks the diagonal elements of the first lag coefficients toward zero, reflecting the limited persistence of income growth rates. Out-of-sample predictive accuracy is assessed by reserving the last 100 observations (corresponding to 25 years) for forecast evaluation, leaving $T = 289$ for model estimation. We use 20,000 draws for every model and discard the first 10% as burn-in.

5.1 Results specific to the Matrix Dirichlet Process Prior

The Matrix Dirichlet Process prior is implemented with conjugate Gamma(1, 1) priors on the hyperparameters α and β . The prior on the coefficient matrix is centered at 0.5 for the first lag's diagonal, and zero for all other coefficients, with a variance of 0.25 for the first lag coefficients, and 0.05 for the second lag. Under this specification, the posterior draws yield an average of 10 clusters (standard deviation 3). Figure B.1 in the Supplemental Appendix displays the pairwise

⁵Data were obtained from <https://www.bea.gov/data/income-saving/personal-income-by-state> on November 30, 2023.

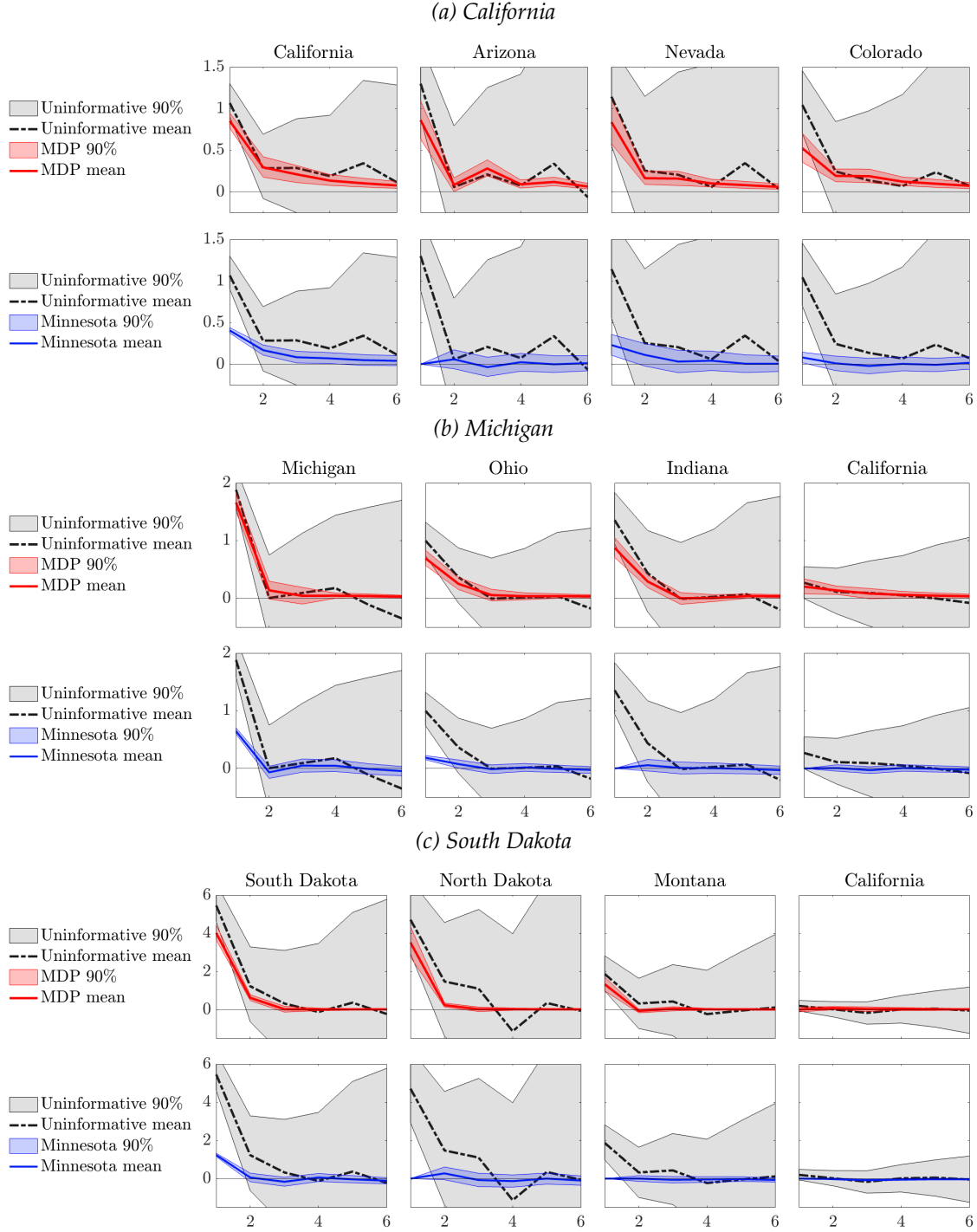
clustering probabilities among states across columns; that is, it shows the likelihood that the effects of state i on other states are clustered with those of state k , with states ordered approximately by geographic region. Notably, there is a distinct cluster comprising Idaho, Utah, Colorado, and Arizona, as well as a larger block encompassing much of the Midwest. In the Northeast, New Jersey and Connecticut are frequently clustered together. By contrast, some states—such as Missouri, New York, and (West) Virginia—are rarely grouped with others, suggesting their estimated parameters and dynamics are distinct from those of other states.

5.2 Comparing Generalized Impulse Response Functions

Figure 4 presents selected model-implied impulse response functions and corresponding 90% credible intervals. We use generalized impulse responses, following Pesaran and Shin (1998), because the model does not admit a natural variable ordering. The analysis considers shocks originating in South Dakota, Michigan, and California, selected to represent economies of differing size and regional importance. South Dakota, a relatively small economy, is expected to generate limited propagation, primarily to neighboring states such as North Dakota. Michigan, with a medium-sized economy, is anticipated to produce spillovers within the Midwest due to regional linkages. California, as the largest state economy, is included to assess the potential for widespread national spillovers.

The posterior mean estimates under the uninformative prior are consistent with the intuition outlined above. However, the associated credible intervals widen rapidly with the horizon, limiting inference at longer lags. The Matrix Dirichlet Process prior produces posterior means that closely track those of the uninformative prior, but with substantially narrower credible intervals as a result of the clustering structure. In each posterior draw, clustering reduces the number of regression coefficients from pN^2 to approximately $pN + 10p$, where $p = 2$ is the number of lags, N is the number of variables, and the average number of clusters is 10. Recall that the diagonal elements are unrestricted. This reduction in dimensionality largely accounts for the improvement in interval precision. The Minnesota prior also yields narrower intervals, but primarily through strong shrinkage toward zero, attenuating economically meaningful relationships; for example, under the Minnesota prior, spillovers from South Dakota to North Dakota and from California to Arizona are close to zero.

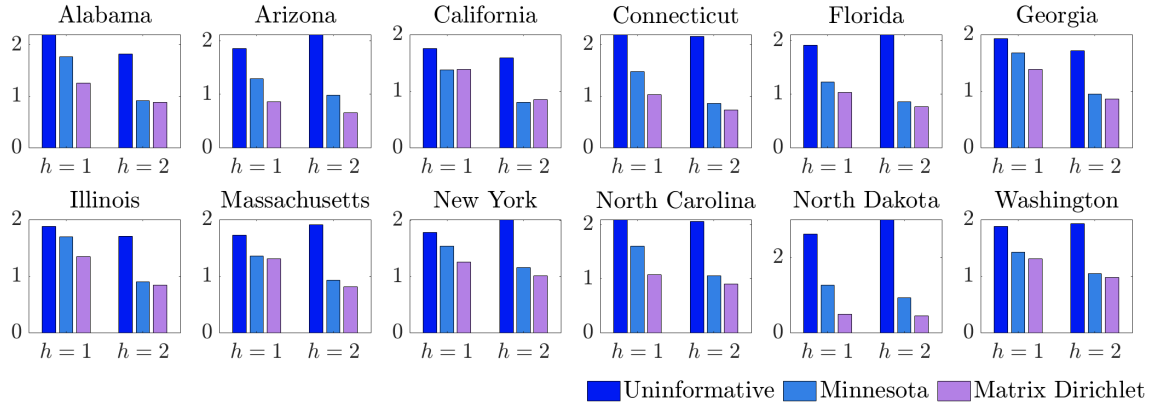
Figure 4: Selected Generalized Impulse Response Functions (IRFs). Each IRF shows the propagation of a shock to the panel state to selected different states (columns). Each panel presents the IRFs under the Matrix Dirichlet Process prior (top row, red) and the Minnesota prior (bottom row, blue), compared to the uninformative prior. Shaded areas show 90% credible intervals.



5.3 Forecasting Performance

Lastly, this application evaluates the forecasting performance of the state-level BVAR(2) estimated under three different priors. The model is first estimated using data through 100 quarters prior to the forecast period (i.e., without a rolling window). With the estimated coefficients, forecasts are then generated recursively for the subsequent 100 quarters. Figure 5 reports the root mean squared forecast errors (RMSFE) for $h = 1, 2$ step-ahead forecasts, normalized by the standard deviation of each series, for a selected set of states. Appendix Figure B.2 presents the corresponding results for all states and for additional horizons (up to $h = 4$).

Figure 5: Selected Root Mean Squared Forecast Errors for $h = 1, 2$ step ahead forecasts for Uninformative Prior, Minnesota Prior and Matrix Dirichlet Process prior for out-of-sample forecasts over the last 25 years. RMSFE divided by standard deviation of each respective series.



For almost all states, the Matrix Dirichlet Process prior yields the lowest RMSFE across all horizons. The only exception is California – shown in Figure 5 – where the Minnesota prior performs marginally better. In some cases, the improvements relative to both the uninformative and Minnesota priors are substantial; for example, at $h = 1$, the RMSFE for North Dakota under the Matrix Dirichlet Process prior is less than half that under the Minnesota prior and one-fifth the RMSFE under the uninformative prior. The differences between the Minnesota and Matrix Dirichlet Process priors diminish at longer horizons, as the model is estimated in growth rates and both priors strongly shrink higher-order lag coefficients toward zero, as can be seen in Figure B.2 in the Appendix, where we show horizons $h = 3$ and 4.

6 Second Application: Monetary Policy Shocks

The previous application involved a panel VAR with the same variable across many different states, lending itself naturally to two-dimensional clustering. We now turn to a less structured setting: a medium-scale VAR of U.S. macroeconomic and financial variables. This allows us to evaluate the method’s performance in high-dimensional systems absent such panel structure and to demonstrate its usefulness for structural analysis. In particular, this application studies impulse responses to monetary policy shocks, where credible interpretation typically depends on the identification strategy.

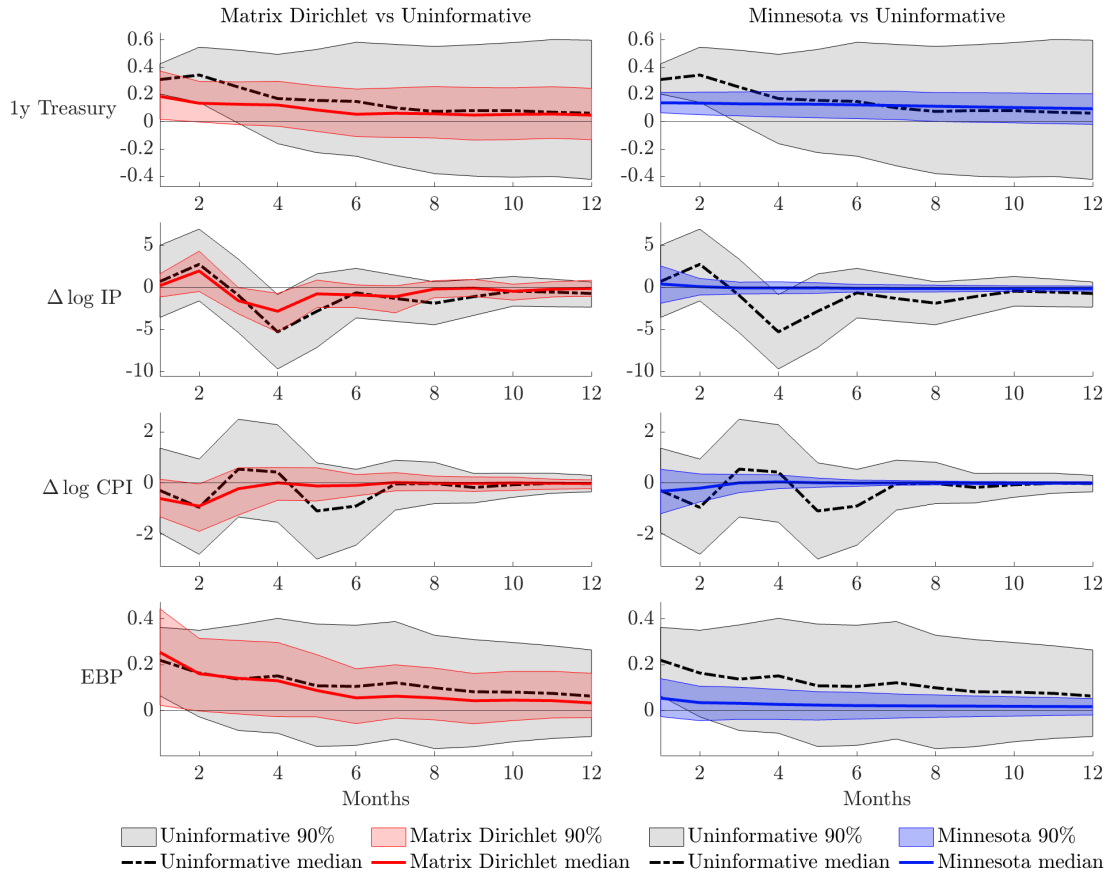
The application follows Plagborg-Møller and Wolf (2021), who show that a proxy variable, ordered first in a Cholesky decomposition, yields impulse responses with a structural interpretation. While large Bayesian VARs have previously been used to study monetary policy (e.g., Bańbura et al., 2010; Crump et al., 2021), they required internal identification schemes, such as recursive orderings or long-run restrictions (Blanchard and Quah, 1989). Recent work has moved toward identification via the use of external instruments, which offer greater empirical credibility (Stock and Watson, 2018), though the applications have so far been limited to small systems, likely in part due to the noisier estimates such an approach produces.⁶ The MDP approach enables us to extend the use of external instruments to high-dimensional VARs, and, in addition, is attractive in connection with the long-standing discussion on the importance of including all variables from the Federal Reserve’s information set (Sims, 1992, Bauer and Swanson, 2023) to avoid omitted variable bias.

We begin by replicating the five-variable VAR analysis in Plagborg-Møller and Wolf (2021), using monthly data on four macroeconomic variables from January 1990 to June 2012 ($T = 270$) and the external instrument for monetary policy shocks constructed by Gertler and Karadi (2015) from high-frequency financial data. The macroeconomic variables — industrial production growth, CPI inflation, the 1-year Treasury yield, and the Excess Bond Premium (EBP) — match those in Gertler and Karadi (2015). Figure 6 presents results from a VAR(4) specification. Under

⁶The use of external instruments with large-dimensional BVARs is especially tricky with Minnesota priors because the instruments can be quite noisy/weak and therefore their coefficients too often get shrunk to zero.

the uninformative prior, the impulse responses replicate the benchmark findings in Plagborg-Møller and Wolf (2021). The Matrix Dirichlet Process prior delivers similar shapes but sharper inference, with the inflation response negative and significant at the 90% level. In contrast, the Minnesota prior, with hyperparameter selected as in Bańbura et al. (2010), shrinks the responses too aggressively, pulling them toward zero.

Figure 6: Comparing impulse responses to a Monetary Policy shock using the Gertler and Karadi (2015) instrument.



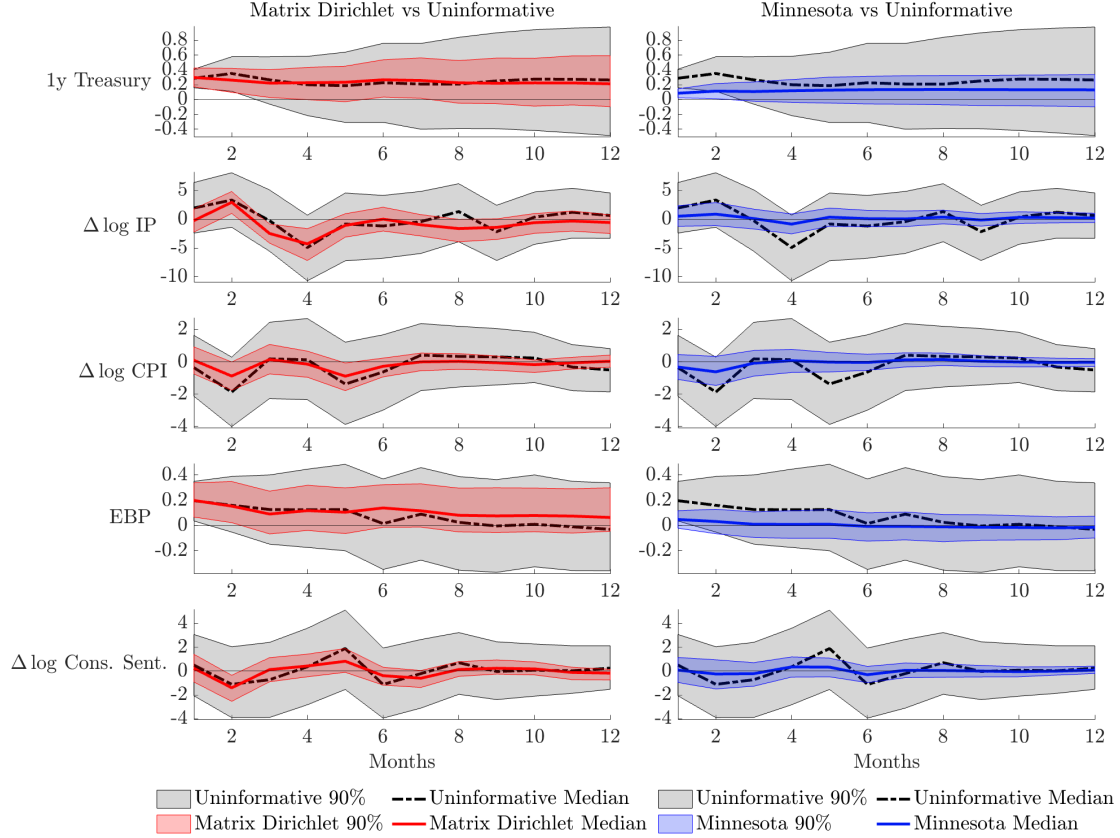
Notes: Impulse response functions from a five-variable VAR(4), compared to using an uninformative prior. The left panel presents IRFs using the Matrix Dirichlet Process prior; the right panel using the Minnesota prior. Shaded areas show 90% credible intervals.

Next, we extend the analysis to an 18-variable VAR(4), using 13 additional time series obtained from the FRED-MD dataset (McCracken and Ng, 2016)⁷. The added variables are a set of real-activity measures (retail sales, building permits, the inventories-to-sales ratio, the civilian unemployment rate), three producer price index (PPI) components (final consumer goods, crude materials, and metals), and a range of financial variables (the BAA–AAA corporate credit spread, the VIX, the S&P 500 index, the S&P dividend yield, the USD/CHF exchange rate, and consumer sentiment). The financial and sentiment variables are forward-looking variables included to control for the potential Federal Reserve information effect (Bauer and Swanson, 2023), and the Producer Price Indices are in response to Sims (1992).

To assess the effect of different priors, Figure 7 compares impulse responses under the uninformative, Matrix Dirichlet Process, and Minnesota priors, focusing on the same four variables as before, i.e., one-year treasury rate, industrial production, CPI inflation, and the excess bond premium, alongside a forward-looking variable: consumer sentiment. For all five variables shown in Figure 7, the uninformative prior yields wide credible intervals that mostly include zero throughout the horizon, except for the excess bond premium at horizon 1. By clustering coefficients on variables with similar dynamic responses, the Matrix Dirichlet Process prior produces narrower intervals that exclude zero at selected horizons, while largely preserving the overall shape and magnitude of the impulse responses. For example, we find significant negative responses in consumer sentiment following an unexpected positive monetary policy shock. In contrast, the Minnesota prior, which shrinks all off-diagonal coefficients toward zero, tends to further attenuate the responses: while intervals are narrower, they uniformly include zero. Figure 8 presents results for a set of financial variables, showing that the Matrix Dirichlet Process prior finds a negative effect on the Swiss-USD exchange rate, while the Minnesota prior again shrinks the effect to zero. Appendix Figure C.1 presents results for the remaining variables, confirming the same patterns.

⁷Data (vintage 2025-05) obtained from <https://www.stlouisfed.org/research/economists/mccracken/fred-databases> on June 24, 2025.

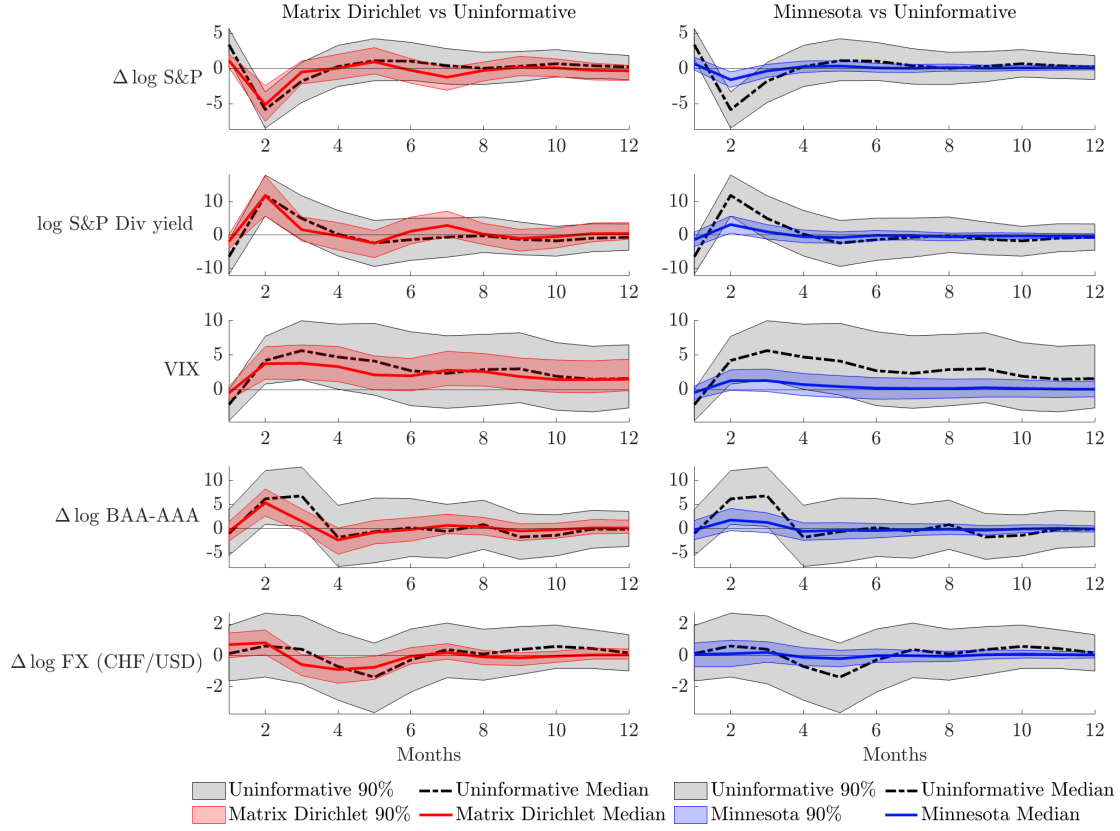
Figure 7: Selected impulse responses to a monetary policy shock in an 18-variable VAR(4) model, comparing uninformative prior to the Matrix Dirichlet Process prior (left, red) and the Minnesota prior (right, blue)



7 Third Application: Uncertainty Shocks and PPI Spillovers

The macroeconomic effects of uncertainty shocks have been analyzed by, among others, Bloom (2009), Carriero, Mumtaz, Theodoridis, and Theophilopoulou (2015), Leduc and Liu (2016), Basu and Bundick (2017), and Fasani and Rossi (2018). Empirical evidence indicates that increased uncertainty – often proxied by the volatility index (VIX) – tends to lower prices, typically identified by placing the VIX or another volatility proxy first in the Cholesky ordering of a VAR. In this application, we wish to analyze how uncertainty shocks propagate, with particular emphasis on the channels through which these shocks affect sector-level prices across the supply-chain network.

Figure 8: Impulse Responses of the financial variables to a monetary policy shock in the 18-variable VAR(4) model of the U.S. economy



Network theory motivates the modeling of shock transmission along inter-industry linkages, leading to the use of structures such as the Network-VAR (NVAR) (Mlikota, 2023) or dynamic factor models (Foerster et al., 2011) to achieve dimension reduction. However, these approaches impose strong restrictions, as spillovers may originate from multiple overlapping networks (e.g., financial linkages in addition to production networks), and sectoral price responses can be heterogeneous due to factors such as buffers or working capital constraints. An unrestricted VAR offers more flexibility, but its high dimensionality can result in imprecise estimates. Matrix Dirichlet Process priors allow for flexible, data-driven parameter restrictions, improving inference and reducing estimation noise.

7.1 Data and implementation details

As our proxy for uncertainty, we use the VIX provided in the FRED-MD data set, as in the previous section. Industry-level monthly producer price indices (PPI) are from the Bureau of Labor Statistics (2000–2019), with $T = 232$, and log-transformed. We restrict attention to sectors with available data from 2000 onward, resulting in a PPI panel of 24 sectors. An overview of sectors is given in Table D.1 in the Appendix. Together with the VIX, this yields a 25-variable VAR.

The flexibility of the MDP prior allows us to cluster only the coefficients associated with the PPI series, while leaving coefficients related to the VIX (i.e., the first column and row) unrestricted. Cluster assignments for the restricted coefficients follow $\alpha_{ih} \sim \text{Beta}(1, 3)$ and $\gamma_{jh} \sim \text{Beta}(1, 3)$. We employ conjugate priors for the VAR lag coefficients and Jeffreys' prior for the error covariance matrix, as detailed in the methodology section. The prior mean for the first lag's diagonal is set to 0.5, while all other coefficients are centered at zero with variance 0.01.⁸ We use four lags, generate 20,000 posterior draws, discard the first 20% as burn-in, and base inference on the remaining samples.

For comparison, the model is also estimated under an uninformative prior and a conjugate Minnesota-like prior, where the first lag's diagonal elements have prior mean 1 and higher-order lags have prior mean zero. The shrinkage parameter is selected following Bańbura et al. (2010).

7.2 Results specific to the Matrix Dirichlet Process prior

The posterior estimates indicate an average of ten clusters, reflecting substantial dimension reduction. In the unrestricted model, with N sectors and the VIX index, there are $(N + 1)^2 p$ parameters to estimate, of which $N^2 p$ pertain to the production network. By leaving the first row and column unrestricted to avoid imposing restrictions on the direct impact of the uncertainty shock, the clustering prior reduces the number of network-related parameters from $N^2 p$ to $N p$ for the unrestricted diagonals and $10 p$ for the clustered off-diagonal parameters on average (i.e., from 2,304 to 96+40), yielding a far more parsimonious specification.

⁸This is a relatively uninformative prior for a panel VAR: with 95% probability, all off-diagonal elements are approximately between ± 0.2 .

Figure 9: Average posterior probabilities that parameters are clustered together. Left: average by columns (i.e., how often the effect of sector j on other sectors is clustered with the effect of sector k on other sectors). Right: average by rows (i.e., how often the effect of other sectors on sector i is clustered with the effect of other sectors on sector k).

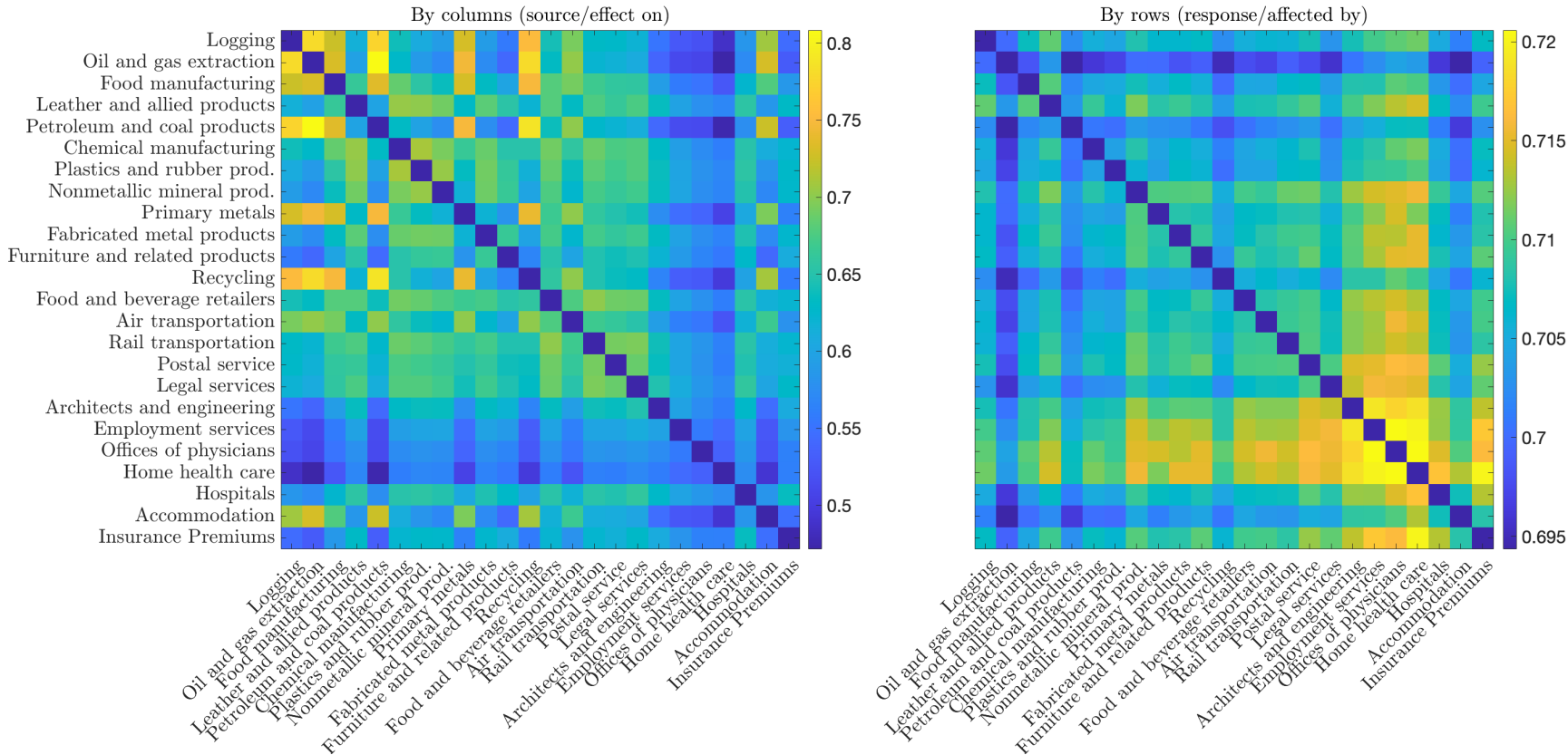


Figure 10: Histogram of parameters in largest cluster (mean (s.e.) in red)

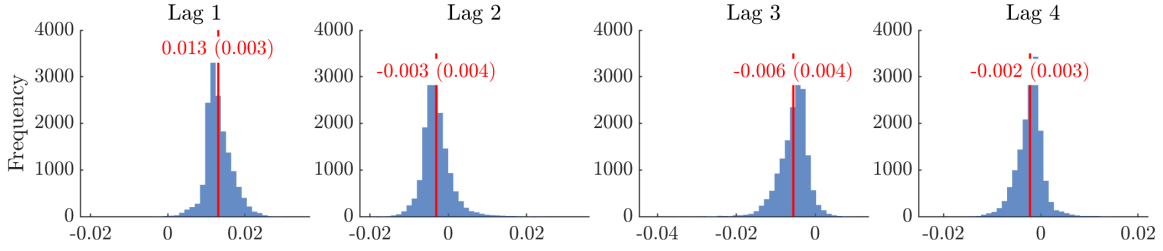


Figure 9 summarizes the resulting cluster structure. Because the full $N^2 \times N^2$ coincidence matrix of all potential parameter pairs (i, j) is too large to visualize, we report averages of the posterior clustering probabilities across rows and columns. Column averages reveal that upstream sectors (e.g., logging, oil and gas extraction, petroleum and coal products manufacturing, primary metal manufacturing) are frequently clustered together, indicating these sectors have similar effects *on other sectors*. Row averages show that service sectors (e.g., legal services, architectural and engineering services, employment services) tend to cluster together, implying similarity in how other sectors affect *them*. Greater dispersion is observed in the column averages, reflecting greater heterogeneity in how sectors affect others than in how they are affected.

Figure 10 displays the parameter distribution for the largest cluster in each draw, illustrating that typical dependence is positive at the first lag, followed by mean reversion in subsequent lags.

7.3 Impulse Response Functions to an Uncertainty Shock

Following the empirical approaches of Bloom (2009) and Basu and Bundick (2017), we order the volatility index first, followed by all PPI series, and compute impulse response functions using a Cholesky decomposition. Figures 11a and 11b display impulse responses to a one-standard deviation uncertainty shock for 10 selected sectors; results for the other 14 sectors are provided in Appendix Figure D.1. We do not visualize impulse responses obtained under the uninformative prior, as their credible intervals are approximately three times wider than those of the other approaches. Instead, we focus on comparisons with the Minnesota prior.

To investigate how the uncertainty shock propagates through the supply-chain network, we decompose the total effect (i.e., the impulse response) into a direct effect – defined as the response when network propagation is shut down by setting all network-related off-diagonal coefficients to zero – and an indirect effect, which captures additional price changes due to network spillovers. In Figures 11a and 11b, the dotted lines represent the direct effect, while the solid lines represent the total effect. The difference between the two reflects the contribution of network propagation.

The standard Minnesota prior is poorly suited for network decompositions, as it shrinks both the direct effects of the VIX on individual PPI series and the network spillover effects toward zero. Consequently, most direct effects under the Minnesota prior (right panels of Figures 11a and 11b) are close to zero.

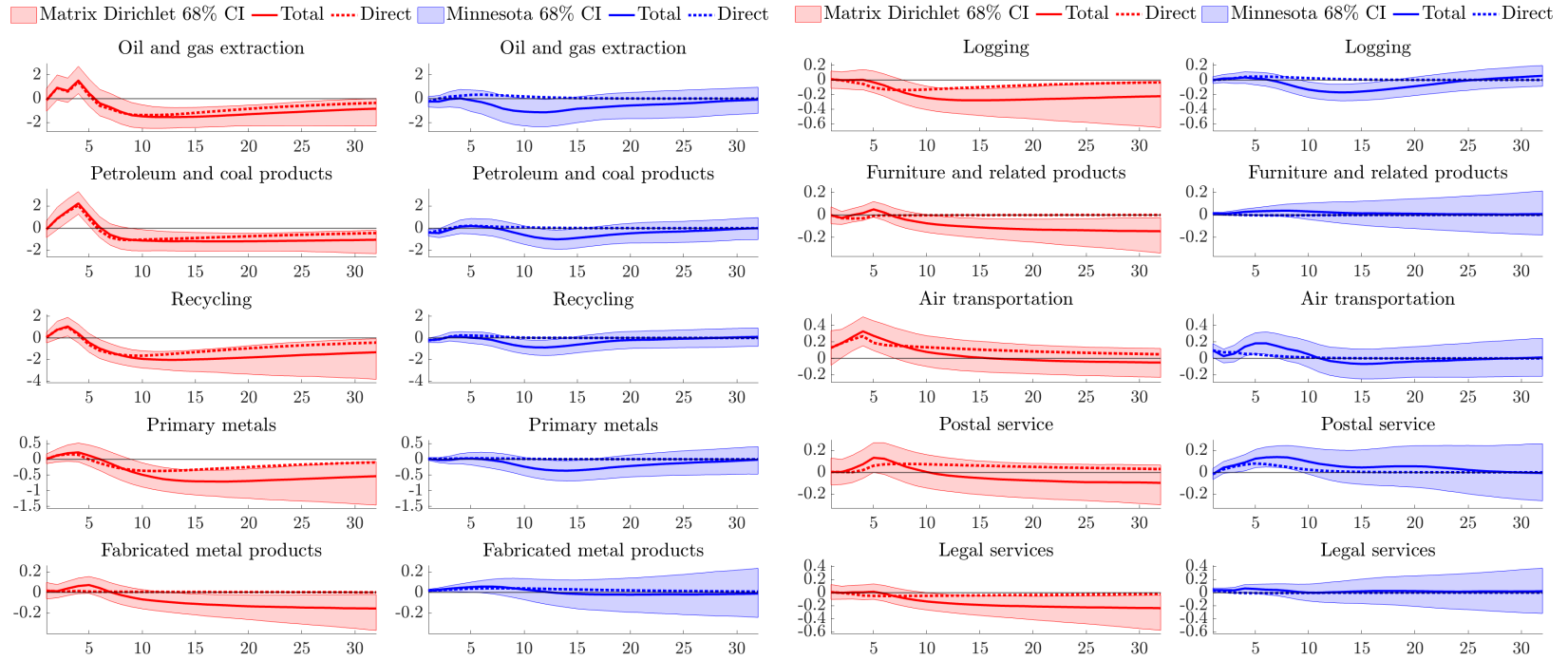
In contrast, the Matrix Dirichlet Process prior reveals a more economically plausible pattern. Upstream sectors exhibit large direct effects and relatively small indirect effects; see, for example, the impulse responses for oil and gas extraction, petroleum and coal products, and recycling in Figure 11a. Moving downstream, to primary metals, indirect network effects become more prominent, while for even more downstream sectors, such as fabricated metal products, the direct effect is nearly zero and most of the price response is attributable to network propagation. A similar pattern emerges in Figure 11b: logging, an upstream sector, displays a larger direct effect, whereas for furniture manufacturing, the effect of an uncertainty shock on that sector's PPI is largely driven by network effects.

Several general patterns are apparent. First, prices initially rise in response to the uncertainty shock, particularly for oil and gas, petroleum and coal, and air transportation. The postal service, which is highly exposed to air transportation costs, also exhibits large initial upward price pressure. After approximately eight months or less, most price indices turn negative, with the exception of air transportation, which remains elevated until roughly one and a half years after the shock. Second, across all sectors, the network contribution to prices is negative: after about eight months, the total effect is consistently lower than the direct effect. Third, the PPI of service sectors, except for postal service, is almost entirely driven by network effects; the direct effect is close to zero, and, driven by the second observation, service-related PPI tends to decline.

Figure 11: Impulse Response Functions (IRFs) to an Uncertainty Shock, 10 selected sectors. Solid lines show the IRFs; dotted lines the direct effect (i.e., the response when network propagation is shut down by setting network-related off-diagonal coefficients to zero). Shaded areas show 68% credible intervals. Panel (a) shows the IRFs of selected upstream and manufacturing sectors while panel (b) shows the IRFs of some additional manufacturing and services sectors). For each panel, the left (red) figures show results under the Matrix Dirichlet Process prior; the right (blue) figures under the Minnesota prior.

(a) Selected upstream and manufacturing sectors

(b) Selected manufacturing and services



8 Conclusion

This paper proposes the use of the Matrix Dirichlet Process (MDP) prior for estimating large Bayesian VARs. The theoretical motivation for this prior comes from the observation that large VARs have poor finite-sample properties, and, in particular, can exhibit substantial second-order bias, which can be mitigated by imposing equality restrictions. However, incorrectly specified restrictions may introduce misspecification bias. The MDP prior balances this trade-off by identifying and imposing equality constraints that are well-supported by the data. As a result, the MDP prior can reduce both variance and second-order bias in settings with large N and small T , challenging the conventional view that regularization methods necessarily involve a bias-variance trade-off.

To make the MDP prior computationally feasible in a large Bayesian VAR framework, we develop a computationally efficient sampling algorithm. The MDP prior effectively exploits the sparsity and symmetry often observed in VAR coefficient matrices by clustering coefficients into groups. Zero coefficients are tightly estimated as a single cluster, while clusters of non-zero coefficients are identified and estimated with greater precision by sharing information across parameters. In three empirical applications, we demonstrate that the MDP prior achieves both improved forecasting performance and more interpretable results compared to traditional shrinkage priors, such as the Minnesota prior. These results suggest that the MDP prior provides a promising alternative for modeling large VARs.

Large Bayesian VARs have been extended to incorporate stochastic volatility (Carriero, Clark, and Marcellino, 2016), time variation (Chan, 2023), and structural restrictions (Chan, 2023). While these extensions are beyond the scope of this paper, they can likely be integrated with the MDP prior. Notably, our implementation accommodates other priors on the coefficient matrices and leaves variance-covariance estimation unrestricted, making it compatible with a wide range of Bayesian VAR frameworks.

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A Proof of Theorem 1

Consider a system of known restrictions $R\beta = r$ where R is a $q \times p$ full-rank matrix and r is a $q \times 1$ vector. Denote by β_r the constrained estimator that satisfies $R\beta = r$. In addition, we assume the restriction is true, hence β_0 , the true parameter, also satisfies $R\beta_0 = r$. Introduce Lagrange multiplier λ and define the Lagrangian function:

$$\mathcal{L}(\beta, \lambda) = \frac{1}{2} \psi_T(\beta)' Q \psi_T(\beta) + \lambda'(R\beta - r).$$

with $Q = \bar{H}_1^{-1}$.

Let $J_\psi(\beta) \equiv \nabla_\beta \psi_T(\beta)$ and evaluate the first-order conditions:

$$\nabla_\beta \mathcal{L} = J_\psi(\beta)' Q \psi_T(\beta) + R'\lambda = 0, \quad (\text{A.1})$$

$$\nabla_\lambda \mathcal{L} = R\beta - r = 0. \quad (\text{A.2})$$

As before, $H_1 = J_\psi(\beta_0)$. Expand $\psi_T(\beta)$ around β_0 :

$$\psi_T(\beta_r) = \psi_T(\beta_0) + H_1(\beta_r - \beta_0) + o_p(T^{-1/2}), \quad \beta_r : R\beta_r = r.$$

Substituting this in the first-order condition:

$$H_1' Q [\psi_T(\beta_0) + H_1(\beta_r - \beta_0)] + R'\lambda \approx 0.$$

Denote $B = H_1' Q H_1$, $b = H_1' Q \psi_T(\beta_0)$. Solving gives:

$$\beta_r - \beta_0 = -Q_r b + o_p(T^{-1/2}), \quad (\text{A.3})$$

where:

$$Q_r = B^{-1} - B^{-1} R' (R B^{-1} R')^{-1} R B^{-1}. \quad (\text{A.4})$$

Note

$$a_{-1/2}^u = -Q \psi_T(\beta_0), \quad a_{-1/2}^r = -Q_r H_1' Q \psi_T(\beta_0) = Q_r H_1' a_{-1/2}^u.$$

Let the second-order bias coefficients for the unrestricted and restricted estimators take the standard forms

$$\begin{aligned} a_{-1}^u &= Q \left(V a_{-1/2}^u + \frac{1}{2} \bar{H}_2 [a_{-1/2}^u \otimes a_{-1/2}^u] \right) := Qd, \\ a_{-1}^r &= Q_r \left(V a_{-1/2}^r + \frac{1}{2} \bar{H}_2 [a_{-1/2}^r \otimes a_{-1/2}^r] \right) := Q_r d_r, \end{aligned}$$

where $a_{-1/2}^u$ and $a_{-1/2}^r$ are the leading bias terms for the unrestricted and restricted estimators, respectively. Define

$$\tilde{P}_B = B^{-1} R' (R B^{-1} R')^{-1} R,$$

such that

$$Q_r = (I - \tilde{P}_B) B^{-1}.$$

The difference in second-order bias is

$$\Delta a_{-1} = a_{-1}^u - a_{-1}^r = (Q - Q_r)d + Q_r(d - d_r).$$

Since $a_{-1/2}^r = Q_r H_1' a_{-1/2}^u$, the difference $d - d_r$ lies in the image of $\tilde{P}_B$ and Q_r annihilates this subspace by construction, we have $Q_r(d - d_r) = 0$, so

$$\Delta a_{-1} = (Q - Q_r)d,$$

Since $Q - Q_r$ is positive semi-definite, it follows that introducing the restrictions $R\beta = r$, under the assumptions that they are correct, results in a decrease in the second-order bias a_{-1} .

B First Application: Supplemental Figures

Figure B.1: Posterior Clustering Probabilities (aggregated by column, i.e., effect on others)

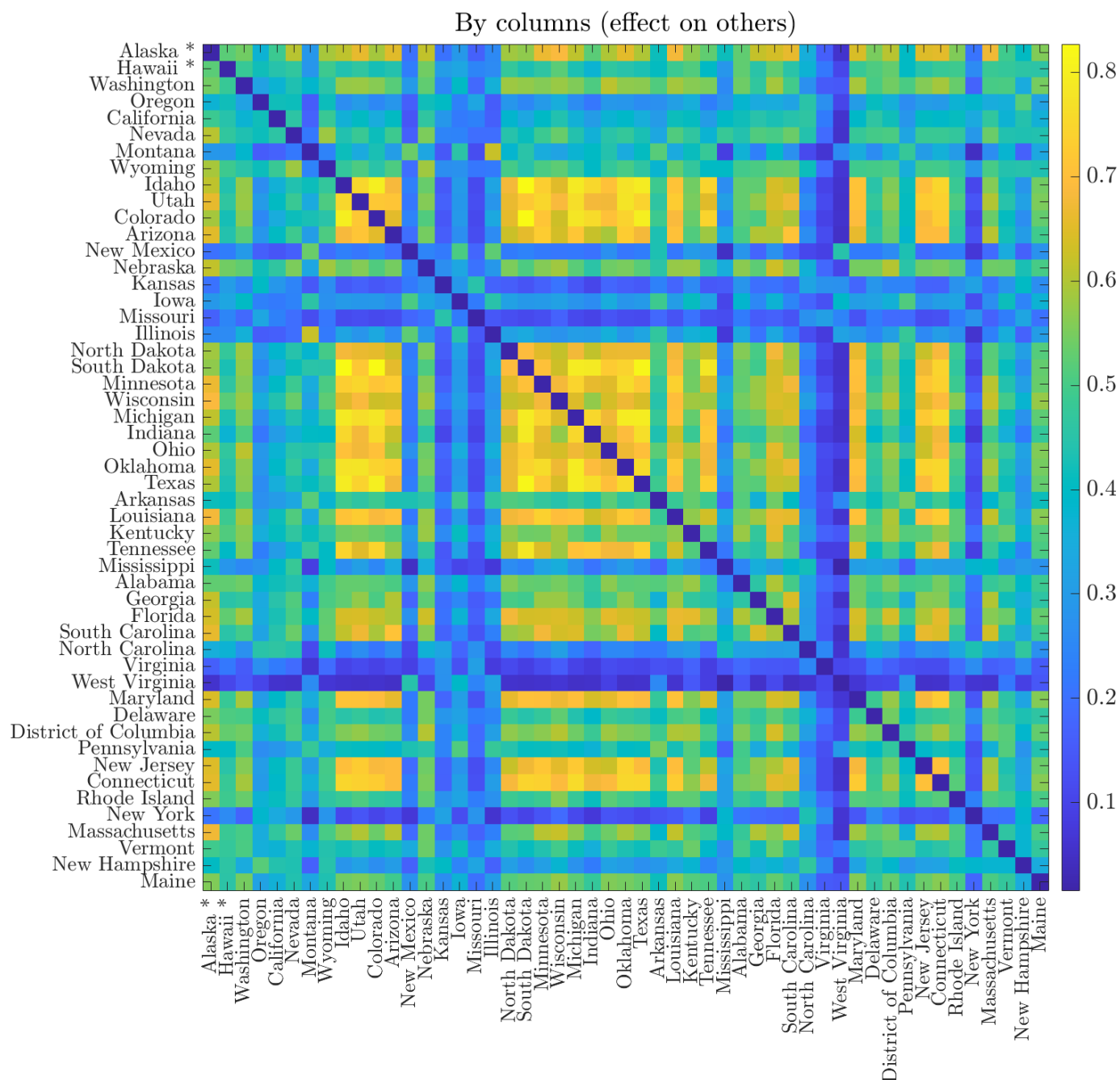
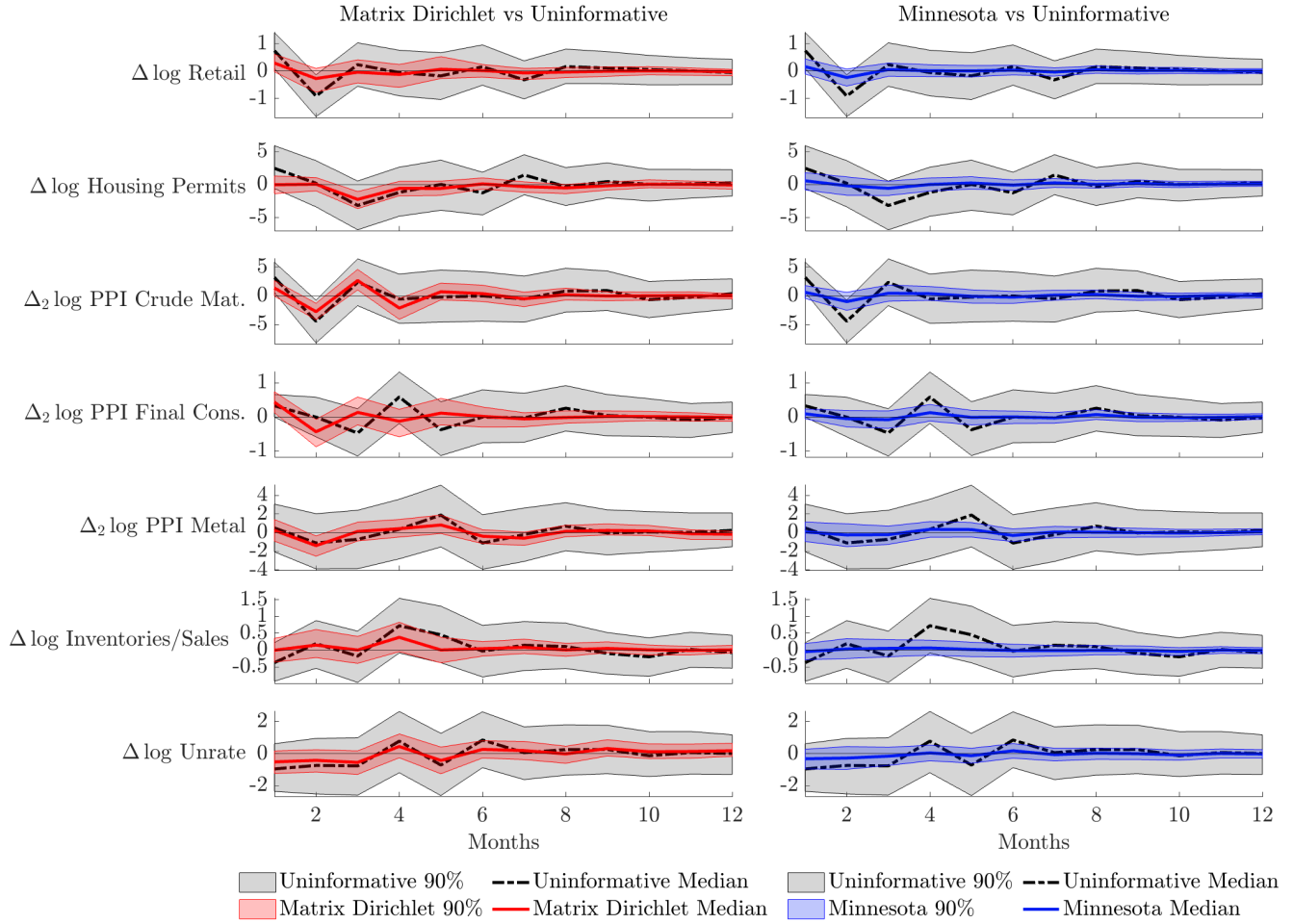


Figure B.2: Root Mean Squared Forecast Errors for $h = 1, 2, 3, 4$ step ahead forecasts for Uninformative Prior, Minnesota Prior and Matrix Dirichlet Process prior for out-of-sample forecasts over the last 25 years. RMSFE divided by standard deviation of each respective series. Model estimated using data up to 100 quarters ago (no rolling window).



C Second Application: Supplemental Figures

Figure C.1: More Impulse Responses for 18-variable VAR(4) of the U.S. economy



D Third Application: Supplemental Figures

Table D.1: Industry Sectors Included in VAR Analysis

PPI Code	Sector Description
PCU1133—1133—	Logging
PCU211—211—	Oil and Gas Extraction
PCU311—311—	Food Manufacturing
PCU316—316—	Leather and Allied Product Manufacturing
PCU324—324—	Petroleum and Coal Products Manufacturing
PCU325—325—	Chemical Manufacturing
PCU326—326—	Plastics and Rubber Products Manufacturing
PCU327—327—	Nonmetallic Mineral Product Manufacturing
PCU331—331—	Primary Metal Manufacturing
PCU332—332—	Fabricated Metal Product Manufacturing
PCU337—337—	Furniture and Related Product Manufacturing
PCU429930429930	Material Recyclers
PCU445—445—	Food and Beverage Retailers
PCU481—481—	Air Transportation
PCU482—482—	Rail Transportation
PCU491—491—	Postal Service
PCU5411—5411—	Legal Services
PCU5413—5413—	Architectural, Engineering, and Related Services
PCU5613—5613—	Employment Services
PCU6211—6211—	Offices of Physicians
PCU6216—6216—	Home Health Care Services
PCU622—622—	Hospitals
PCU721—721—	Accommodation
PCU924126924126	Premiums for Property and Casualty Insurance

Figure D.1: Impulse Response Functions to Uncertainty Shock. The left panel (red) uses the Matrix Dirichlet Process prior, the right panel (blue) the Minnesota prior.

